#1 · Theory and History

How did technical analysis evolve from manual record keeping into a testable discipline?

Answer

Analysts first recorded prices, highs, lows, and volume by hand to compare market behavior. Published services and computers expanded the number of series and tests, while modern practice added data controls, out of sample checks, costs, and monitoring. The purpose stayed the same: turn observations into evidence that can be rejected.

#2 · Theory and History

What major ideas are associated with Charles Dow and later Dow Theory?

Answer

Dow work with stock averages framed primary, secondary, and minor movements, confirmation between industrial and transportation activity, and the use of price and volume to judge market health. Later writers organized these observations into Dow Theory. Dow did not leave a complete trading system under that name.

#3 · Theory and History

How does the Wyckoff framework organize a market campaign?

Answer

It organizes ranges into accumulation, markup, distribution, and markdown. Tests, springs, and signs of strength or weakness help the analyst judge how supply is being absorbed or distributed. The claimed phase must be supported by price and volume evidence.

#4 · Theory and History

What is the main benefit and main danger of Elliott Wave analysis?

Answer

The Wave Principle can organize a complex price path into motive and corrective structures and define where the count fails. Its danger is a flexible count created after the move. A useful count must be stated in advance, follow the core rules, and include an invalidation price.

#5 · Theory and History

How should a candidate evaluate a Gann relationship in an applied market decision?

Answer

Gann work combines price, time, geometry, and numerical relationships such as angles, squares, retracements, and calendar observations. The analyst should require a repeatable entry, invalidation rule, and evaluation method. A geometric observation without those elements remains context rather than a complete trading method.

#6 · Theory and History

Why do box size and reversal size matter in point and figure analysis?

Answer

They decide which price changes appear on the chart. Changing either setting can change the columns, signal count, support and resistance readings, and price projection. Construction therefore shapes the evidence and must be stated before interpretation.

#7 · Theory and History

What evidence standard does modern systematic practice add to traditional chart analysis?

Answer

A rule must have a clear definition, an adequate sample, relevant market conditions, transaction costs, and out of sample evidence. The analyst also states what evidence would reject the rule. Familiar names or attractive historical charts are not enough.

#8 · Theory and History

What does the premise that price discounts available information mean, and when does it weaken?

Answer

The current trade reflects the choices buyers and sellers make after processing what they know and expect. It does not mean the price is always correct or that everyone has the same information. The premise weakens when information is hidden, the market is closed, or a quote is stale.

#9 · Theory and History

Why may a price trend persist, and why is persistence not a promise?

Answer

Slow information flow, gradual adoption, portfolio flows, risk limits, and reactions to recent prices can keep supply or demand moving in one direction. A trend remains temporary and probabilistic. A moving average can describe the trend but cannot prove the next return will have the same sign.

#10 · Theory and History

What must an analyst test before treating a recurring chart pattern as useful evidence?

Answer

Similar fear, greed, hesitation, imitation, and forced liquidation can leave similar footprints, but similarity is not identity. The analyst compares the present pattern with the volume, duration, location, and confirmation in the historical sample. The features that gave the pattern value must still be present.

#11 · Theory and History

Which observable market behaviors provide evidence of supply and demand?

Answer

Price, volume, gaps, range width, failed moves, and recovery speed show how buyers and sellers respond. Expanding volume on a breakout differs from falling volume, and a pullback that holds above a breakout differs from one that cuts through it. Several agreeing behaviors provide stronger evidence than one chart shape.

#12 · Theory and History

If the first stock's volume falls from 3.0 million to 1.0 million while the second stock's rises from 2.5 million to 3.5 million, which has stronger volume confirmation and what are the ratios?

Answer

The second stock's ratio is 3.5 / 2.5 = 1.4, while the first stock's ratio is 1.0 / 3.0 = one third. The second stock therefore shows stronger evidence of active demand. The evidence still needs price structure and later confirmation before it proves a lasting advance.

#13 · Theory and History

What market conditions make technical evidence more reliable?

Answer

Technical methods work best in liquid, freely traded markets with enough independent participants, accessible information, meaningful volume, and prices that can move with supply and demand. Depth and competition help one order or a fixed price from distorting the record. Thin markets, wide spreads, and no trade periods reduce reliability.

#14 · Theory and History

A stock closes at 102 and then trades at 51 after a two for one split. How should the apparent 50 percent decline be corrected?

Answer

The raw ratio is 51 / 102 = 0.50, but the split doubles the share count and halves the price per share. The earlier price should be adjusted to 51 for comparison, making the corrected ratio 51 / 51 = 1.00. The raw move is a capitalization change rather than a market collapse.

#15 · Theory and History

Which data checks are needed before a technical pattern or backtest is trusted?

Answer

The series must use comparable split and dividend adjustments, complete volume, consistent timestamps, and consistent session boundaries. Duplicate prints, missing bars, stale quotes, mixed reporting conventions, and survivorship bias can create false patterns. The data must represent the same economic series through time.

#16 · Theory and History

How should technical evidence fit inside a broader investment decision?

Answer

Technical evidence should define a setup, trigger, and invalidation within a stated market context. Fundamental, liquidity, portfolio, and risk evidence can confirm, delay, or reduce the action. Position size is set separately because a good chart cannot repair excessive risk.

#17 · Theory and History

Why can a model with a positive historical result still fail in live trading?

Answer

The model may be overfit to a sample, sensitive to nearby parameter values, or suited only to a market regime that has ended. Execution costs, slippage, crowding, and unavailable post signal prices can also erase the result. Live monitoring must compare current conditions with the model's domain.

#18 · Theory and History

A model completes 12 round trips with a cost of 0.15 percent on each entry and exit. If gross edge is 2.8 percentage points, what is the estimated net result?

Answer

Total cost is 12 × 2 × 0.15 = 3.6 percentage points. The estimated net result is 2.8 - 3.6 = -0.8 percentage point before taxes and additional slippage. Positive gross signals do not make a process profitable after execution costs.

#19 · Theory and History

What does beta measure in a benchmark return model?

Answer

Beta measures the sensitivity of an investment's returns to benchmark returns. It is the covariance of the two return series divided by benchmark variance.

#20 · Theory and History

How is Jensen's alpha calculated from portfolio return?

Answer

Subtract the beta adjusted expected return from the actual portfolio return. The expected return is the riskless return plus beta times the benchmark return minus the riskless return.

#21 · Theory and History

Why can positive active return occur with zero alpha?

Answer

Active return is simply portfolio return minus benchmark return. A portfolio with beta above 1 can gain relative to the benchmark during a rally even when its return matches the amount explained by that higher exposure.

#22 · Theory and History

A fund returns 14%, the benchmark returns 10%, the riskless return is 2%, and beta is 1.25. What are alpha and active return?

Answer

Expected return is 2% plus 1.25 times 8%, or 12%. Alpha is 2 percentage points and active return is 4 percentage points.

#23 · Theory and History

What information do the weak, semi strong, and strong forms of EMH treat as reflected in price?

Answer

The weak form concerns past market data, the semi strong form concerns public information, and the strong form includes private information. Each form claims that dependable excess profit from that information set is difficult to obtain.

#24 · Theory and History

How does the adaptive markets hypothesis describe market efficiency?

Answer

Efficiency changes as investors make mistakes, learn, compete, and adapt. An edge can weaken as capital arrives, so a model must be monitored across changing market conditions.

#25 · Theory and History

What role do investment horizons play in the fractal market hypothesis?

Answer

Participants with different horizons can make a market more stable. Instability can rise when many participants crowd into the same horizon and react in a similar way.

#26 · Theory and History

What must a technical analyst prove before calling a temporary supply imbalance an opportunity?

Answer

The analyst must show that the imbalance is being absorbed and that stronger price behavior follows. Light selling volume, recovery of a key level, or a failed move can be evidence, but the setup still needs a defined invalidation point.

#27 · Theory and History

How can a sharp price move caused by forced selling be distinguished from a rational repricing?

Answer

Forced flow often reverses after volume falls and key levels are reclaimed. A rational repricing tends to hold or extend as participants update value. Random noise produces no dependable sequence.

#28 · Theory and History

What different jobs do fundamental and technical analysts perform in fusion analysis?

Answer

The fundamental analyst estimates business value, drivers, and catalysts. The technical analyst assesses current supply and demand, trend confirmation, timing, and invalidation.

#29 · Theory and History

How can two analysts with opposite views still use fusion analysis productively?

Answer

They can align their time horizon and benchmark, identify the evidence that would change each view, and build a conditional plan. A smaller position, a wait for stabilization, or a structural stop can combine their findings.

#30 · Theory and History

What is anchoring bias in a market decision?

Answer

Anchoring gives too much weight to an initial reference such as a purchase price, prior high, or round number. The chart can support the idea of clustered orders, but supply and demand evidence must confirm it.

#31 · Theory and History

How does availability bias affect the estimate of a rare market event?

Answer

A vivid recent event feels more likely, while an event absent from recent memory may feel nearly impossible. The analyst should use a longer history and measure the setup's base rate.

#32 · Theory and History

How does representativeness bias create a chart reading error?

Answer

A vivid or incomplete move may be treated as proof of a repeating pattern. The analyst should require a completed structure and compare the setup with its historical success rate in the current regime.

#33 · Theory and History

How can confirmation bias weaken a technical review?

Answer

The analyst may count supporting signals and ignore weakening breadth, poor volume, or failed resistance. Writing the invalidation rule before entry forces contrary evidence into the decision.

#34 · Theory and History

What is the disposition effect and how can it harm position management?

Answer

It is a loss aversion pattern in which investors sell winners early and hold losers to avoid realizing a loss. A predefined stop and a separate profit management rule reduce this emotional bias.

#35 · Theory and History

What are the five stages of a market bubble?

Answer

The stages are displacement, boom, euphoria, profit taking, and panic. They describe a possible sequence, but timing varies and a real market can skip or compress a stage.

#36 · Theory and History

What chart evidence is most useful near the transition from euphoria to profit taking?

Answer

Watch for a parabolic price slope, failed breakouts, weakening breadth, lower quality leadership, heavier downside volume, and rising realized volatility. A single overbought reading is not enough.

#37 · Classical Techniques

What job should the highest timeframe serve in a top down chart process?

Answer

It defines the broad market environment, the major trend, and the important invalidation level. Lower timeframes should refine structure and execution rather than replace that higher timeframe conclusion.

#38 · Classical Techniques

How should an analyst choose the chart intervals for a trade?

Answer

Choose intervals after defining the expected holding period. Use charts separated enough to answer different questions about strategy, structure, and tactical execution.

#39 · Classical Techniques

What does a daily pullback inside a weekly uptrend usually represent?

Answer

It can be a normal countertrend move while the weekly structure remains intact. It becomes evidence of a larger reversal only when key higher timeframe support fails with follow through.

#40 · Classical Techniques

Why is timeframe shopping a process error?

Answer

The analyst changes to a longer timeframe after a trade moves against the original plan. This uses hindsight to avoid a loss rather than testing a rule chosen before the trade.

#41 · Classical Techniques

Why do equal time bars contain unequal information?

Answer

A fixed clock interval can contain very different amounts of volume, trades, and price movement. A quiet bar and an event driven bar may have the same duration but not the same information content.

#42 · Classical Techniques

How do volume bars and time bars sample the market differently?

Answer

Time bars form after a fixed clock interval, so their count is stable but activity varies. Volume bars form after a fixed amount of volume, so bar count varies with participation.

#43 · Classical Techniques

Why can including premarket trade change a daily candle?

Answer

The added trades can change the open, high, low, close, and volume relative to a regular session only candle. The analyst must use one session definition consistently and state it.

#44 · Classical Techniques

What are three possible causes of a missing intraday bar?

Answer

The market may have been closed, the data feed may have failed, or no trades may have occurred. These causes have different analytical meanings, so the analyst should identify the cause before repairing data.

#45 · Classical Techniques

Why is filling a missing price bar with the prior close dangerous?

Answer

It invents a flat price and a false no trade interval. Moving averages, volume, range, and volatility can all become biased by the fabricated observation.

#46 · Classical Techniques

Why should two daily series use a common information cutoff in a short term comparison?

Answer

Different closes can place market moving news in different bars. The mismatch can create false lead and lag patterns even when both daily charts look normal on their own.

#47 · Classical Techniques

Why is a dual price axis overlay a weak leadership comparison?

Answer

The analyst can change each axis and change where the lines appear to cross. Rebase both series or use a ratio line so relative performance has a defined scale.

#48 · Classical Techniques

What does a rising ratio of asset A to benchmark B establish?

Answer

It establishes that A is outperforming B over that interval. It does not prove that A is rising or that A has positive absolute momentum.

#49 · Classical Techniques

An asset stays at 100 while its benchmark falls from 200 to 180. What is the new ratio and relative result from the old ratio of 0.50?

Answer

The new ratio is 100 divided by 180, or 0.5556. That is an 11.1 percent rise, so the asset is outperforming even though its own price is unchanged.

#50 · Classical Techniques

What is a Time Price Opportunity in Market Profile?

Answer

It marks that a price range traded during a defined time bracket. Stacked TPOs form a profile that shows how much time the auction spent at each price.

#51 · Classical Techniques

What is the point of control in a Market Profile?

Answer

It is the price with the greatest number of TPOs in the profile, so it is the most accepted price for that sample. It is a location of acceptance, not automatically a price target.

#52 · Classical Techniques

How is the value area described in common Market Profile teaching?

Answer

It is the price band containing a chosen majority share of TPOs, often about 70 percent. The value area high and value area low mark its upper and lower edges.

#53 · Classical Techniques

How do TPO profiles and volume profiles measure price differently?

Answer

A TPO profile measures time spent at price. A volume profile measures traded quantity at price, so the two profiles can have different points of control and shapes.

#54 · Classical Techniques

What does sustained acceptance above prior value suggest?

Answer

It suggests the auction is finding trade at higher prices and may be discovering a new value area. Repeated activity and a profile built above prior value support that conclusion.

#55 · Classical Techniques

What does a move above value followed by a quick return into value suggest?

Answer

It suggests rejection of higher prices because trade above value did not persist. A single print above value is weaker evidence than repeated acceptance.

#56 · Classical Techniques

How does arithmetic scale space price changes?

Answer

Equal vertical distances represent equal dollar or point changes. A move from 20 to 40 has the same chart height as a move from 80 to 100.

#57 · Classical Techniques

How does logarithmic scale space price changes?

Answer

Equal vertical distances represent equal percentage changes. Doubling from 20 to 40 has the same height as doubling from 50 to 100.

#58 · Classical Techniques

Why does a constant percentage growth rate appear straight on a logarithmic chart?

Answer

A constant compound growth rate adds equal log changes over equal periods. Logarithmic spacing converts those equal proportional changes into equal visual distances.

#59 · Classical Techniques

A price rises from 20 to 40, then from 80 to 100. Why does the first move matter more on a log chart?

Answer

The first move is a 100 percent gain while the second is a 25 percent gain. Equal 20 point gains occupy very different heights when percentage change is the vertical unit.

#60 · Classical Techniques

What does trendline slope mean on an arithmetic chart?

Answer

It describes the change in dollars or points per bar. A steep arithmetic line means rapid price point change, not necessarily rapid compound return.

#61 · Classical Techniques

What does trendline slope mean on a logarithmic chart?

Answer

It describes proportional or compound change per bar. A steep log line means rapid percentage change and should not be described as a dollar slope.

#62 · Classical Techniques

When is logarithmic scale especially useful?

Answer

It is useful for long histories, large price ranges, and comparisons where percentage change matters more than equal dollar steps. The scale should match the analytical question and the units of the target.

#63 · Classical Techniques

How is an additive price target calculated after a range breakout?

Answer

Measure the range in price points and add that distance to the breakout level. A range from 50 to 60 produces an upside additive target of 60 plus 10, or 70.

#64 · Classical Techniques

How is a proportional upside target calculated after a range from 50 to 60?

Answer

The range is a 20 percent gain, so repeating it from 60 gives 60 times 1.20, or 72. This method preserves percentage distance rather than point distance.

#65 · Classical Techniques

Why do a 10 point additive target of 70 and a proportional target of 72 differ?

Answer

The additive method repeats 10 raw points. The proportional method repeats a 20 percent change and compounds it from a higher base.

#66 · Classical Techniques

Which projection method is visually consistent with a logarithmic chart?

Answer

A proportional projection is consistent because equal log distances represent equal percentage changes. Mixing raw point projections with log distance readings breaks that consistency.

#67 · Classical Techniques

Why does a fixed point stop represent changing proportional risk as price rises?

Answer

The same point amount is a larger share of a low price and a smaller share of a high price. Use the scale and risk unit that match how the position is actually sized.

#68 · Classical Techniques

What does a line chart hide that an OHLC or candlestick chart shows?

Answer

A close only line hides the intrabar high, low, open, and sequence. It can make volatility and rejection within a session look smaller than they were.

#69 · Classical Techniques

Why must long term adjusted price history be handled carefully?

Answer

Unadjusted splits and dividends can create false gaps, while futures rolls need a defined continuous series. Adjustment preserves economic continuity but changes the displayed price history.

#70 · Classical Techniques

What should an analyst verify before trusting a multi timeframe chart conclusion?

Answer

Verify the instrument, session, time zone, bar rule, missing data treatment, scale, and projection method. The signal is only as reliable as the chart construction behind it.

#71 · Classical Techniques

What is the usual sequence of the four price and volume phases?

Answer

The cycle is usually described as accumulation, markup, distribution, and markdown. These phases are descriptive and often overlap, so their boundaries may be clear only after the fact.

#72 · Classical Techniques

Which price and volume clues point to accumulation?

Answer

Price stops making lower lows while selling pressure becomes less urgent. Rallies may begin to carry more volume, and pullbacks may become smaller or quieter.

#73 · Classical Techniques

What does healthy volume usually look like during markup?

Answer

Volume tends to expand on advances and contract on orderly pullbacks as participation broadens. Repeated heavy volume without price progress is a warning that supply is meeting demand.

#74 · Classical Techniques

Which evidence suggests distribution near a high?

Answer

Price progress slows while volume remains active and leaders begin to weaken. Failed rallies, heavier down days, and deteriorating breadth together provide stronger evidence than one heavy session.

#75 · Classical Techniques

What changes during markdown?

Answer

Price develops lower highs and lower lows, with heavier volume on declines and lighter volume on rebounds. A late volume climax shows forced activity, but a bottom still needs evidence that price is responding.

#76 · Classical Techniques

Why should volume be compared with its recent average?

Answer

A raw volume number has little meaning without a baseline. Relative volume shows whether current participation is above or below the normal activity for that market and period.

#77 · Classical Techniques

How does volume usually confirm an uptrend?

Answer

Expanding volume on advances shows that more participants are accepting higher prices. Orderly pullbacks on quieter volume show less urgent selling.

#78 · Classical Techniques

How should an analyst treat a breakout on below average volume?

Answer

The breakout is less well supported and should be treated as unconfirmed. A later close above resistance or a successful retest can strengthen the evidence, while a quick return into the range is a warning.

#79 · Classical Techniques

What distinguishes constructive and suspect periods of low volume?

Answer

A quiet pullback to support in an uptrend can be constructive. A quiet rally back toward broken resistance in a downtrend can show a lack of demand.

#80 · Classical Techniques

Why is climax volume not an automatic reversal signal?

Answer

A climax shows forced or unusually urgent activity, not a completed change in direction. Reversal evidence requires price to stop extending and then show a changed response, such as a recovery from the low.

#81 · Classical Techniques

How do you calculate relative volume for a 750,000 share session against a 1.2 million share average?

Answer

Divide 750,000 by 1,200,000. The result is 0.625, so the session is 62.5 percent of the recent average.

#82 · Classical Techniques

What does the advance decline line measure?

Answer

It is a running total of advancing issues minus declining issues. Its direction and divergence from price matter more than its absolute level because the starting value is arbitrary.

#83 · Classical Techniques

How is the advance decline ratio calculated?

Answer

Divide the number of advancing issues by the number of declining issues. A value of 1.0 means the two counts are equal.

#84 · Classical Techniques

Why use up and down volume as well as issue counts?

Answer

Issue counts show how many securities participate, while volume shows how much trading supports those moves. A small number of heavily traded stocks can dominate volume breadth.

#85 · Classical Techniques

What is the TRIN formula?

Answer

TRIN equals advancing issues divided by declining issues, then divided by advancing volume divided by declining volume. It combines issue breadth with volume breadth.

#86 · Classical Techniques

What does a TRIN reading below 1.0 normally indicate?

Answer

Advancing volume is relatively stronger than the count of advancing issues. This usually shows more forceful activity on the upside, but the value should be judged against its own history.

#87 · Classical Techniques

What does the percentage of stocks above a moving average reveal?

Answer

It estimates how many securities are participating in the prevailing trend. A rising percentage indicates improving participation, while a falling percentage indicates narrowing strength.

#88 · Classical Techniques

What does the McClellan Oscillator measure?

Answer

It measures the short term momentum of net advances by comparing a shorter EMA with a longer EMA. It is a breadth oscillator, not a direct count of bullish and bearish securities.

#89 · Classical Techniques

How should a lower high in breadth during a price high be interpreted?

Answer

It suggests that fewer securities are supporting the advance and that risk is rising. It does not prove that price has reversed, so price confirmation remains necessary.

#90 · Classical Techniques

What does a rising relative strength line show about leadership?

Answer

It shows that the security or sector is outperforming its benchmark. The absolute price can still be falling, so relative strength and price direction must be considered separately.

#91 · Classical Techniques

How can a qualitative review of market internals stay testable?

Answer

Describe observable conditions such as which sectors lead, whether breakouts hold, and how median stocks behave. Avoid vague conclusions and state which future price or breadth evidence would invalidate the view.

#92 · Classical Techniques

What limits the reliability of breadth indicators?

Answer

Index membership changes, unequal security weights, exchange traded funds, new listings, and market structure can distort the measures. Breadth divergences can also persist for a long time in a strong trend.

#93 · Classical Techniques

Why is insider buying often treated differently from insider selling?

Answer

Insider sales are often routine responses to compensation, taxes, or diversification. Discretionary open market purchases can provide a stronger signal because they are less commonly forced.

#94 · Classical Techniques

Which details make insider buying more meaningful?

Answer

Repeated purchases by several senior insiders are more useful than one small purchase. Open market buying after a decline and buying near the executive's own cost can add weight.

#95 · Classical Techniques

How do you calculate short interest as a percentage of float?

Answer

Divide the number of shares sold short by the number of shares available in the float. If 12 million shares are short and the float is 100 million, short interest is 12 percent.

#96 · Classical Techniques

How do you calculate days to cover from 12 million short shares and 3 million average daily volume?

Answer

Divide short interest by average daily volume. The result is 12 million divided by 3 million, or 4 days to cover.

#97 · Classical Techniques

Why does high short interest not guarantee a squeeze?

Answer

Short sellers may be correct, and the reported position is delayed. A squeeze becomes more plausible when a catalyst forces buying, liquidity is limited, and price begins to rise against the short position.

#98 · Classical Techniques

How should the bull bear spread in a sentiment survey be used?

Answer

Use it to identify a crowded opinion and track changes through time. A level alone has limited value because bullish or bearish sentiment can remain extreme while price continues to move.

#99 · Classical Techniques

When does contrary opinion become stronger evidence?

Answer

It becomes stronger when sentiment reaches an extreme and price action shows that the crowd is being forced to change. An extreme reading without price confirmation is only a warning.

#100 · Classical Techniques

How do price and futures open interest change interpretation together?

Answer

Rising price with rising open interest suggests new positions support the move. Rising price with falling open interest often reflects short covering, while falling price with rising open interest suggests new short positions.

#101 · Classical Techniques

How should the Commitments of Traders report be interpreted?

Answer

It reports commercial, noncommercial, and nonreportable positions. Nonreportable positions are a residual whose trader count and commercial or noncommercial classification are unknown. Use the report to assess crowding and position changes, not as a direct prediction.

#102 · Classical Techniques

What does a high put call ratio suggest, and what limits that inference?

Answer

A high ratio can show defensive sentiment or crowded hedging, which may be contrarian near an extreme. Option type, open interest versus volume, and the users of equity and index options can change the meaning.

#103 · Classical Techniques

How can options volatility provide sentiment evidence?

Answer

Rising implied volatility during a decline shows increased demand for protection. Falling implied volatility during an advance suggests reduced fear, but volatility readings must be compared with their own history.

#104 · Classical Techniques

What must be established before a visual shape is treated as a usable chart pattern?

Answer

The analyst should mark the boundaries, identify the relevant swing, assess volume and trend context, and define completion. The pattern label should follow the evidence rather than replace it.

#105 · Classical Techniques

How is the downside objective of a head and shoulders top calculated?

Answer

Measure from the head to the neckline, then subtract that height from the neckline breakout level. An inverse head and shoulders adds the height to the neckline.

#106 · Classical Techniques

What supports a head and shoulders completion?

Answer

A close below the neckline with expanding volume supports a downside completion. A right shoulder that is similar in scale to the left shoulder adds structural quality.

#107 · Classical Techniques

How is a double top price objective commonly projected?

Answer

Measure the height from the support line to the peaks, then project that distance below the support after a confirmed break. A double bottom projects the height upward.

#108 · Classical Techniques

Why does a third resistance test not make a triple top complete by itself?

Answer

The pattern still needs a decisive break of its support or neckline. Without that break, the same price action can remain a rectangle or another range.

#109 · Classical Techniques

How does a rounding bottom usually develop?

Answer

Price decelerates after a decline, flattens, and gradually turns upward. Volume is often quiet near the center and expands during the recovery.

#110 · Classical Techniques

What does an ascending triangle show?

Answer

Rising lows press into a roughly horizontal resistance level. A confirmed close above that ceiling favors continuation of an existing uptrend.

#111 · Classical Techniques

What does a descending triangle show?

Answer

Falling highs press into a roughly horizontal support level. A confirmed close below support raises the risk of a downside continuation.

#112 · Classical Techniques

Does a symmetrical triangle predict breakout direction?

Answer

No. It shows contraction in price and time. The larger trend, the boundary broken, and the confirmation evidence determine the trade interpretation.

#113 · Classical Techniques

How is a flag target measured?

Answer

Measure the sharp move that formed the flagpole, then project that length from the flag breakout. The output is a planning objective, not a guaranteed destination.

#114 · Classical Techniques

How does a pennant differ from a flag?

Answer

A pennant is a small triangle that narrows after a sharp move, while a flag is a small parallelogram that usually slopes against the trend. Both use the prior pole for a measured objective.

#115 · Classical Techniques

Why should a wedge label be treated with caution?

Answer

A wedge can act as a continuation or become part of a reversal, and definitions vary with context. The pattern boundaries and larger trend matter more than the label.

#116 · Classical Techniques

What tactics fit a price still inside a rectangle?

Answer

A mean-reversion trader may buy near support with risk beyond the boundary and target the middle or opposite side. A breakout tactic is used only after price leaves the range and satisfies confirmation.

#117 · Classical Techniques

What confirms the completion of a cup and handle setup?

Answer

A confirmed move through the old rim after a rounded cup and a controlled handle supports completion. A handle that falls back deep into the cup weakens the structure.

#118 · Classical Techniques

What is a measured move?

Answer

It projects a completed swing or pattern height into a future price area from a breakout level. The result is an estimate for planning, not a promise of support or resistance.

#119 · Classical Techniques

How should pattern width be used in target analysis?

Answer

Width mainly describes duration, maturity, and context. A fixed multiplication of width is not a universal price target, so the analyst should state the measurement method being used.

#120 · Classical Techniques

Resistance is 50 and the 3 percent rule is used. What upside close confirms?

Answer

Three percent of 50 is 1.50, so the threshold is 51.50. A close above 51.50 satisfies the stated rule.

#121 · Classical Techniques

Support is 80 and the 3 percent rule is used. What downside close confirms?

Answer

Three percent of 80 is 2.40, so the threshold is 77.60. A close below 77.60 satisfies the stated rule.

#122 · Classical Techniques

Resistance is 48.20 and the rule is one half ATR with ATR at 1.50. What upside threshold applies?

Answer

One half of 1.50 is 0.75, so the threshold is 48.20 plus 0.75, or 48.95. The close basis and ATR lookback should be stated.

#123 · Classical Techniques

How does an ATR confirmation rule differ from a 3 percent rule?

Answer

The ATR rule scales with recent price movement, while the percentage rule scales with price. The ATR buffer adapts to volatility but requires a disclosed lookback and multiple.

#124 · Classical Techniques

Is a 3 percent threshold the same as a 3 percent price objective?

Answer

No. The threshold is the amount beyond a boundary that must be cleared for confirmation. The price objective comes from the pattern height or another stated measurement.

#125 · Classical Techniques

What is a reasonable mean-reversion entry near range support?

Answer

Wait for evidence that support is holding, place risk beyond the range, and target the middle or opposite side. A quiet break of support weakens the mean-reversion premise.

#126 · Classical Techniques

What evidence supports a breakout entry beyond a range boundary?

Answer

A qualified close beyond the boundary, expanding volume, and follow-through support the breakout. A throwback that holds the old boundary can offer a second entry.

#127 · Classical Techniques

What is a throwback after a breakout?

Answer

It is a quick return to the broken boundary, which may act as new support or resistance. Holding the retest supports the breakout, while a close back inside the range weakens it.

#128 · Classical Techniques

Why can a breakaway pullback improve a trade plan?

Answer

It can place entry nearer to a structural invalidation level and improve the reward-to-risk calculation. The cost is that the pullback may not arrive or may become a failed breakout.

#129 · Classical Techniques

What market behavior most clearly weakens a confirmed upside breakout?

Answer

A close back inside the original pattern after the breakout weakens the thesis. A subsequent break of the last important swing low gives stronger evidence of failure.

#130 · Classical Techniques

What is a common gap?

Answer

It occurs in a quiet or choppy market and often appears on limited volume. It can fill quickly because the gap does not show strong new conviction.

#131 · Classical Techniques

What is a breakaway gap?

Answer

It appears as price leaves a consolidation or pattern, often with strong volume. It is associated with a new directional move rather than ordinary range noise.

#132 · Classical Techniques

What is a runaway gap?

Answer

It appears during an active trend and is also called a measuring gap. It can be used within a measured-move framework because it reflects acceleration in the established direction.

#133 · Classical Techniques

What is an exhaustion gap?

Answer

It appears late in a move, often with high volume, and may be followed by weak continuation and a partial or full fill. Location and follow-through matter more than size alone.

#134 · Classical Techniques

How should volume be used when classifying a gap?

Answer

Strong volume supports a breakaway or directional interpretation, while quiet volume fits a common gap. Volume must be combined with the trend, location, and subsequent closes.

#135 · Classical Techniques

Where can a stop be placed for a failed range breakout?

Answer

It can go beyond the broken boundary, beyond the swing that formed the setup, or at a disclosed ATR buffer outside the structure. Position size should follow from the entry-to-stop distance.

#136 · Classical Techniques

How does an invalidation condition differ from a stop order?

Answer

Invalidation is the evidence that changes the trade thesis, such as a close back inside a pattern. A stop is the order that limits the loss and may sit beyond invalidation to avoid ordinary noise.

#137 · Classical Techniques

What elements belong in a classical pattern trade plan?

Answer

The plan should state pattern evidence, entry trigger, confirmation rule, invalidation, stop, target, and reward-to-risk. It should also define how to respond to a retest, failure, or first target.

#138 · Classical Techniques

Which four prices define one candlestick?

Answer

A candlestick uses the open, high, low, and close for one selected interval. The body spans the open and close, while the shadows reach the high and low.

#139 · Classical Techniques

What does a long candlestick body show?

Answer

A long body shows that the open and close finished far apart. It indicates a large net price change during the interval, but it does not by itself identify whether the move will continue.

#140 · Classical Techniques

What information does a candlestick shadow provide?

Answer

A shadow shows that price reached an extreme and then moved back before the close. Its meaning depends on the trend, the location, and the volume during the session.

#141 · Classical Techniques

How is close location within a candle range calculated?

Answer

Subtract the low from the close, then divide by the high minus the low. A close near the upper end shows stronger intraperiod demand than a close near the lower end.

#142 · Classical Techniques

Why can the same candle shape mean different things on two charts?

Answer

The charts may use different intervals, trends, support zones, or volume conditions. A setup on a daily chart can be a small pause on a weekly chart.

#143 · Classical Techniques

What structure defines a hammer?

Answer

A hammer has a small body near the top of its range and a long lower shadow. It gains reversal meaning after a decline, especially at support with confirming follow through.

#144 · Classical Techniques

How does a hanging man differ from a hammer?

Answer

The two candles have the same basic shape. A hanging man occurs after an advance, while a hammer occurs after a decline, so market context changes the interpretation.

#145 · Classical Techniques

What does a shooting star show?

Answer

A shooting star has a small body near the low of its range and a long upper shadow after an advance. It shows that a rally met supply and failed to hold its gain.

#146 · Classical Techniques

How is an inverted hammer interpreted?

Answer

An inverted hammer has a small body near the low and a long upper shadow after a decline. It can show an early demand attempt, but a later close above the candle is needed for stronger confirmation.

#147 · Classical Techniques

What does a doji reveal?

Answer

A doji has an open and close that are equal or nearly equal. It shows indecision because neither buyers nor sellers held a meaningful closing advantage during the interval.

#148 · Classical Techniques

How do long legged, dragonfly, and gravestone dojis differ?

Answer

A long legged doji has long shadows on both sides. A dragonfly has mainly a lower shadow, while a gravestone has mainly an upper shadow.

#149 · Classical Techniques

What is a spinning top candle?

Answer

A spinning top has a small body and shadows on both sides. It records a pause or disagreement in control, but it is usually less distinctive than a doji.

#150 · Classical Techniques

What does a marubozu indicate?

Answer

A marubozu has a long body with little or no shadow. A bullish version closes near its high and a bearish version closes near its low, showing a sustained move during the interval.

#151 · Classical Techniques

What is a bullish engulfing pattern?

Answer

A bullish candle follows a bearish candle and its real body covers the prior real body. After a decline, it shows buyers overcoming the previous session's selling range.

#152 · Classical Techniques

What is a bearish engulfing pattern?

Answer

A bearish candle follows a bullish candle and its real body covers the prior real body. After an advance, it shows sellers overcoming the previous session's buying range.

#153 · Classical Techniques

What does a harami show?

Answer

A harami has a small body contained within the prior large body. It shows that the prior move paused, but it is often weaker evidence than an engulfing reversal.

#154 · Classical Techniques

What is the rule for a piercing pattern?

Answer

After a decline, a bullish candle opens below or near the prior low and closes above the midpoint of the prior bearish body. Deeper penetration gives stronger visual evidence.

#155 · Classical Techniques

What is the rule for dark cloud cover?

Answer

After an advance, a bearish candle opens above or near the prior high and closes below the midpoint of the prior bullish body. The pattern suggests that sellers have seized control.

#156 · Classical Techniques

What three candles form a morning star?

Answer

A long bearish candle, a small body candle or doji, then a long bullish candle that closes well into the first body. The sequence appears after a decline.

#157 · Classical Techniques

What three candles form an evening star?

Answer

A long bullish candle, a small body candle or doji, then a long bearish candle that closes well into the first body. The sequence appears after an advance.

#158 · Classical Techniques

What do tweezer tops and tweezer bottoms show?

Answer

They show two candles testing nearly the same high or low. The repeated test is meaningful when it occurs at support or resistance and later price action confirms it.

#159 · Classical Techniques

Why does volume matter for a candlestick signal?

Answer

Volume shows how much participation supported the candle's move. A reversal on volume well above its recent average is more credible than the same shape on unusually light activity.

#160 · Classical Techniques

Why does support or resistance change a candle pattern's meaning?

Answer

The zone identifies a price where buyers or sellers previously made decisions. A reversal candle there can show a new test of that decision, while the same shape in the middle of a range may be noise.

#161 · Classical Techniques

What is the difference between a complete and a confirmed candlestick pattern?

Answer

A complete pattern has all its required candles after the final close. A confirmed pattern has later price action that supports the intended reversal or continuation.

#162 · Classical Techniques

What does an incomplete candlestick pattern mean in real time?

Answer

Some required candles or price relationships are still missing. The analyst should identify what the market must do to finish or cancel the setup instead of assuming the pattern already exists.

#163 · Classical Techniques

How should an imperfect candlestick pattern be handled?

Answer

Judge trend, location, control transfer, volume, and the next close before trusting the label. Reduce confidence or wait for a cleaner trigger when the pattern's parts conflict.

#164 · Classical Techniques

What invalidates a bullish candlestick setup?

Answer

A decisive close below the signal low or the support zone weakens the setup and can trigger the planned stop. A brief shadow through the level is different from a confirmed close beyond it.

#165 · Classical Techniques

What structure defines an abandoned baby?

Answer

An abandoned baby uses three candles with a doji isolated by gaps between the first and third candles. It is a strict star formation and the gaps are central evidence.

#166 · Classical Techniques

What structure defines an island reversal?

Answer

Price gaps away from a trend, trades for one or more candles in an isolated area, then gaps back toward the original price area. The return gap completes the reversal structure.

#167 · Classical Techniques

How does an abandoned baby differ from an island reversal?

Answer

An abandoned baby has three candles and an isolated doji. An island reversal can contain any number of candles and is defined more broadly by an isolated price area created by gaps on both sides.

#168 · Classical Techniques

What sequence should an analyst use to integrate candle evidence?

Answer

Establish the larger trend, identify support or resistance, assess the candle pattern's completeness, compare volume, and require later confirmation. Then set the entry, invalidation, and position size.

#169 · Classical Techniques

Why should overlapping candlestick signals not be counted as independent evidence?

Answer

Several candle features can describe the same session and the same price recovery. Confidence should increase only when the candle agrees with evidence from a different source, such as trend, volume, or a classical chart pattern.

#170 · Classical Techniques

How should risk be planned for a candlestick trade?

Answer

Define the market level that would prove the signal wrong, measure entry risk to that level, and size the position from the planned account risk. The candle name does not replace a structural stop.

#171 · Classical Techniques

How do X and O columns and time work on a point and figure chart?

Answer

An X column records rising prices and an O column records falling prices. Time does not advance the axis by itself, so a new mark appears only when price reaches the next box threshold or completes the specified reversal.

#172 · Classical Techniques

How is a point and figure reversal threshold calculated?

Answer

Multiply the box size by the number of reversal boxes. A one dollar box with a three box reversal requires a three dollar move against the current column before a new column can begin.

#173 · Classical Techniques

Why does scale matter in point and figure analysis?

Answer

Arithmetic boxes show equal dollar changes, while logarithmic boxes show equal proportional changes. The scale should match the price range and the intended comparison.

#174 · Classical Techniques

What does congestion represent on a point and figure chart?

Answer

Congestion is a horizontal band where X and O columns overlap. It marks a price area where control repeatedly changed and provides the base used in horizontal counts.

#175 · Classical Techniques

How is a vertical point and figure count applied in this chapter?

Answer

Count the boxes in the breakout column, multiply by box value and reversal amount, then project from the column low for X or the column high for O. Eight one dollar boxes with a three box reversal give 24 points, so an X column low at 75 targets 99.

#176 · Classical Techniques

How does a proportional point and figure count differ from an arithmetic count?

Answer

A fixed dollar count adds or subtracts dollars. A percentage box count compounds proportional moves on a log chart, so the result depends on the starting price.

#177 · Classical Techniques

What does a horizontal point and figure count measure?

Answer

It measures the number of columns across a congestion area, then applies the stated box value and reversal amount. A wider base can support a larger count but does not guarantee a larger move.

#178 · Classical Techniques

How are P and F point and figure symbols used in relative strength analysis?

Answer

A relative strength ratio divides a security or sector by a benchmark. Rising P columns can show improving relative performance, while F columns can show weaker performance, but the chart legend must confirm the symbols.

#179 · Classical Techniques

What is technical momentum?

Answer

Momentum measures the amount of price change over a chosen lookback. It can be expressed as a price difference, a percentage rate of change, or another ratio.

#180 · Classical Techniques

How is rate of change calculated?

Answer

Divide the current price by the price from the chosen lookback and subtract one. A move from 50 to 54 gives 54 divided by 50 minus 1, or 8 percent.

#181 · Classical Techniques

What is the difference between momentum and acceleration?

Answer

Momentum measures price change from a reference point, while acceleration measures the change in momentum. Price can rise while momentum falls if the rate of advance slows.

#182 · Classical Techniques

Why are overbought readings different in trends and ranges?

Answer

In a range, an overbought reading can warn of resistance and mean reversion. In a strong uptrend, the reading can remain high for many bars as demand persists.

#183 · Classical Techniques

How does momentum typically behave through a directional trend?

Answer

Momentum often rises with price early in the move, may flatten or weaken near the middle, and can lose force before price completes a turn. It can reverse several times before price changes trend.

#184 · Classical Techniques

What is a positive momentum divergence?

Answer

Price makes a lower low while momentum makes a higher low. It suggests that the later decline has less force, but price confirmation is still required.

#185 · Classical Techniques

What is a negative momentum divergence?

Answer

Price makes a higher high while momentum makes a lower high. It shows that the later advance has weaker force, but it does not prove that price has peaked.

#186 · Classical Techniques

How do number and duration affect a divergence?

Answer

Repeated divergences can show persistent loss of force, and a divergence spread across several months can carry more warning than two nearby swings. Neither feature guarantees a price reversal.

#187 · Classical Techniques

Why does the position of a divergence matter?

Answer

A divergence at a major support or resistance zone, or late in an extended trend, has more context than a divergence inside a noisy range. An early warning can still be followed by further trend movement.

#188 · Classical Techniques

What are the three practical steps of a valid momentum signal?

Answer

The indicator changes direction, passes a defined trigger, and receives price confirmation. A turn in the indicator alone is only the first step.

#189 · Classical Techniques

How does a momentum reversal differ from a price reversal?

Answer

A momentum reversal changes the indicator's direction or position. A price reversal changes market structure, such as a lower high followed by a break of the prior low.

#190 · Classical Techniques

What are three ways to recognize a trend change in an oscillator?

Answer

Look for a break in the oscillator's swing pattern, a break of an indicator trendline, or a zero or signal line cross. Faster triggers usually accept more false signals.

#191 · Classical Techniques

Why should an indicator trigger and a price confirmation be separated in a trade plan?

Answer

The indicator may turn before price structure changes. An early entry needs a wider invalidation and smaller risk, while waiting for price confirmation may improve evidence but worsen the entry price.

#192 · Classical Techniques

What does RSI compare?

Answer

RSI compares recent average gains with recent average losses, usually over 14 periods. The 70 and 30 levels are common guides, but trends can keep readings near either edge.

#193 · Classical Techniques

What does the stochastic oscillator measure?

Answer

It measures the latest close relative to the high and low of a lookback period. A common setup is 14 periods with three period smoothing for percent K and percent D.

#194 · Classical Techniques

What are the standard MACD lines and settings?

Answer

With 12, 26, and 9 settings, the MACD line is the 12 period exponential average minus the 26 period average. The signal is a nine period exponential moving average of the MACD line, and the histogram is MACD minus signal.

#195 · Classical Techniques

How do RSI, stochastic, MACD, and rate of change differ in use?

Answer

RSI shows gain loss balance, the stochastic shows close location in a range, MACD shows the spread between fast and slow averages, and rate of change shows direct percentage change.

#196 · Classical Techniques

What was the original purpose of VWAP?

Answer

VWAP was created as an execution benchmark for large institutional orders. It helps compare the average price achieved with the average price paid across the session.

#197 · Classical Techniques

How is VWAP calculated?

Answer

Multiply each trade or bar price by its volume, add those dollar values, and divide by total volume. With bar data, a typical price of high plus low plus close divided by three is often used.

#198 · Classical Techniques

What makes an execution good relative to VWAP?

Answer

A buy is generally favorable when its average price is below VWAP. A sell is generally favorable when its average price is above VWAP, with the result still compared with the stated benchmark and costs.

#199 · Classical Techniques

What does price above VWAP indicate about session control?

Answer

Price above VWAP shows that recent trading is occurring above the volume weighted average cost. It supports a buyer control interpretation, but it does not guarantee continued gains.

#200 · Classical Techniques

How is percent b calculated for Bollinger Bands?

Answer

Subtract the lower band from price, then divide by the upper band minus the lower band. A price at the upper band has a reading of 1.00, while a price halfway between the bands has 0.50.

#201 · Classical Techniques

How does BandWidth differ from percent b?

Answer

BandWidth measures the distance between the bands relative to the middle average. Percent b measures where price sits inside those bands, so it can change even when BandWidth is unchanged.

#202 · Classical Techniques

How do a Bollinger squeeze, bulge, and band walk differ?

Answer

A squeeze is unusually narrow BandWidth and signals low volatility without choosing direction. A bulge is rapid band expansion after volatility rises. A band walk is a series of closes near one band during a strong trend.

#203 · Classical Techniques

What is a bearish two bar reversal at the upper Bollinger Band?

Answer

One bar closes above the upper band and the next closes back inside it. The sequence shows rejection of the expansion and gains weight after an extended rally or at resistance.

#204 · Classical Techniques

What does a market cycle model describe?

Answer

A cycle model describes a recurring rhythm in price, time, or participation. It is a testable expectation rather than a promise that the same move will repeat exactly.

#205 · Classical Techniques

What does cycle amplitude measure?

Answer

Amplitude measures the size of a cycle, often the distance from a trough to a crest or from a crest to a trough. It helps compare the importance of different cycles.

#206 · Classical Techniques

What does a cycle period measure?

Answer

The period is the time between comparable points in a cycle, such as one trough and the next trough. It is the cycle length used in timing projections.

#207 · Classical Techniques

What does cycle phase describe?

Answer

Phase describes where the current date or price sits within the cycle. A known period without known phase does not identify whether the next turn is likely to be a high or a low.

#208 · Classical Techniques

How is a harmonic related to a larger cycle?

Answer

A harmonic has a period that is a whole-number fraction of a larger cycle. The second harmonic of a 40-day cycle has a 20-day period.

#209 · Classical Techniques

What is a nominal cycle used for?

Answer

A nominal cycle is a simplified reference model with a stated period and phase. The analyst compares real market behavior with that model and revises it when amplitude, timing, or shape changes.

#210 · Classical Techniques

What does cycle summation mean?

Answer

Cycle summation treats observed price as a combination of several cycles plus trend and noise. The combined result can strengthen, weaken, or shift the apparent turning date.

#211 · Classical Techniques

How is a dominant cycle different from the longest cycle?

Answer

The dominant cycle is the cycle with the strongest useful amplitude in the study window. A shorter cycle can dominate tactical price swings while a longer cycle defines the broader environment.

#212 · Classical Techniques

How can a spectrogram help find a dominant cycle?

Answer

A spectrogram shows cycle strength across time and period. A strong band marks the active period, and movement of that band shows when the dominant rhythm is changing.

#213 · Classical Techniques

Why is finding a period not enough to time a turn?

Answer

The analyst must also establish phase and amplitude. The same period can place the next high or low on different dates depending on where the current swing sits.

#214 · Classical Techniques

What is left translation?

Answer

Left translation means the peak of a cycle occurs before the midpoint between two troughs. A peak on day 12 of a 30-day cycle is left translated.

#215 · Classical Techniques

What is right translation?

Answer

Right translation means the peak occurs after the midpoint between two troughs. A peak on day 20 of a 30-day cycle is right translated.

#216 · Classical Techniques

How should translation affect a trading decision?

Answer

Right translation is often consistent with upward pressure and left translation with downward pressure, but neither is a standalone signal. Require price confirmation and a clear invalidation level.

#217 · Classical Techniques

What is a cycle inversion?

Answer

An inversion occurs when an expected trough appears as a relative high or an expected crest appears as a relative low. It usually means the simple model is misphased or another cycle is dominating.

#218 · Classical Techniques

What can cause a mid-cycle dip?

Answer

A shorter harmonic can pull price down near the midpoint of a larger nominal cycle. The dip reflects cycle summation and phase offset rather than a universal cycle rule.

#219 · Classical Techniques

What can cause a three-quarter-cycle high?

Answer

A harmonic crest can coincide with the later part of a larger cycle and create a secondary high. The timing depends on the shorter cycle's period and phase.

#220 · Classical Techniques

How can cycle summation explain rounded tops and V-bottoms?

Answer

Rounded turns can occur when nearby cycles spread their turning influence, while V-turns can occur when several cycles reverse together. The shape still needs price and volume confirmation.

#221 · Classical Techniques

How does a centered moving average reduce timing lag?

Answer

A centered average uses data before and after the middle bar, so its value aligns with the center of the window. A trailing average places more weight on older information and lags more.

#222 · Classical Techniques

What is the limitation of a 39-day centered moving average in real time?

Answer

The newest 19 bars do not yet have the future observations required to finish the average. The live portion must be estimated or replaced with a trailing measure.

#223 · Classical Techniques

How is a simple CMA envelope constructed?

Answer

Add and subtract a measured distance from the centered average to create upper and lower bands. The distance can reflect observed half-amplitude or a stated percentage.

#224 · Classical Techniques

What makes a cycle valid trendline useful?

Answer

A valid trendline connects comparable cycle pivots and respects the price behavior between them. It helps show whether the timing structure is intact, but a line break needs wider evidence.

#225 · Classical Techniques

What are two common ways to detrend price data?

Answer

Subtract a regression line from price, or divide price by a moving average and express the result as a percentage. Proportional data are usually better for markets analyzed on a logarithmic scale.

#226 · Classical Techniques

How is a seasonal cycle tested?

Answer

Compare each calendar period with a baseline such as a centered 12-month average or a detrended return. Check stability, dispersion, sample size, and whether the effect survives costs.

#227 · Classical Techniques

What is the difference between finding and phasing a cycle?

Answer

Finding estimates which periods have power, often with spectral analysis or a spectrogram. Phasing estimates where the current move sits within the chosen period and usually uses pivots, envelopes, or trendlines.

#228 · Classical Techniques

What sequence should a comprehensive cycle analysis follow?

Answer

Clean the data, inspect the wider trend, detrend, find candidate periods, test stability and phase, project turns, compare price evidence, and define invalidation. The step from timing forecast to trade requires confirmation.

#229 · Classical Techniques

What is the basic operating idea of the Elliott Wave Principle?

Answer

It proposes that crowd behavior develops in recurring directional and corrective structures across timeframes. A wave count is a probabilistic map with alternate scenarios, not a certainty.

#230 · Classical Techniques

How are motive and corrective Elliott waves usually labeled?

Answer

A motive sequence is commonly labeled 1 through 5. A corrective sequence is commonly labeled A, B, and C, with notation changed to show degree.

#231 · Classical Techniques

What are the core rules for an Elliott impulse?

Answer

Wave 2 cannot retrace more than 100% of wave 1, and wave 3 cannot be the shortest of waves 1, 3, and 5. In a standard cash-market impulse, wave 4 should not overlap wave 1 territory.

#232 · Classical Techniques

What does wave 1 and wave 4 overlap suggest?

Answer

Overlap makes a standard cash-market impulse unlikely. Consider a diagonal pattern or a different wave count instead of keeping a count that breaks the normal rule.

#233 · Classical Techniques

How does an extension differ from a diagonal?

Answer

An extension is a prolonged impulse wave, often wave 3, that subdivides into a smaller five-wave sequence. A diagonal contains five overlapping waves, usually inside converging or expanding boundaries.

#234 · Classical Techniques

How are zigzag, flat, and triangle corrections described?

Answer

A zigzag is a sharp 5-3-5 correction, a flat is often a 3-3-5 sideways correction, and a triangle commonly contains five overlapping 3-wave segments. Their shape affects where the correction may end.

#235 · Classical Techniques

What does alternation mean in corrective waves?

Answer

Nearby corrections such as wave 2 and wave 4 often differ in shape, depth, or duration. Alternation is a guideline for comparing scenarios, not a strict rule.

#236 · Classical Techniques

How are Fibonacci targets calculated for a wave 2 and a wave 3 example?

Answer

If wave 1 rises 20 points, a 0.618 retracement is 12.36 points. From a wave 2 low, a 1.618 extension of the same 20-point length is 32.36 points. Use the result as a zone and require structural confirmation.

#237 · Advanced Techniques

How do a population and a sample differ in a technical-analysis study?

Answer

The population is the complete set of conditions described by the research question. The sample is the measured subset used to make an inference about that population.

#238 · Advanced Techniques

What role does the null hypothesis play in a statistical test?

Answer

The null hypothesis is the baseline claim that the sample is asked to contradict, often stating no effect, no difference, or no positive edge. The evidence either rejects it or fails to reject it at the chosen threshold.

#239 · Advanced Techniques

How should a one-sided alternative hypothesis be used?

Answer

It states a direction before the data are inspected, such as a mean return greater than zero. Choosing the direction after seeing the result uses hindsight and distorts the test.

#240 · Advanced Techniques

What does a test statistic do?

Answer

It converts the sample result into a scale that can be compared with a reference distribution. For a sample mean, it often divides the distance from the hypothesized mean by the standard error.

#241 · Advanced Techniques

What is a critical value?

Answer

A critical value is the cutoff in a reference distribution for the chosen significance level and test direction. A statistic beyond the cutoff falls in the rejection region.

#242 · Advanced Techniques

How is a p-value interpreted?

Answer

It is the probability, assuming the null hypothesis is true, of obtaining a test statistic at least as extreme as the observed value. It is not the probability that the null hypothesis is true.

#243 · Advanced Techniques

What does failing to reject a null hypothesis establish?

Answer

It establishes only that the sample did not provide enough evidence to reject the null at the chosen threshold. It does not prove that the null statement is true.

#244 · Advanced Techniques

When is a two-sided test more suitable than a one-sided test?

Answer

A two-sided test is suitable when the analyst asks whether an effect differs in either direction. A one-sided test is suitable only when the economic question and direction were specified in advance.

#245 · Advanced Techniques

What is a Type I error in a trading-rule test?

Answer

A Type I error is rejecting a true null hypothesis, which is a false positive. The rule is declared useful when no positive edge exists.

#246 · Advanced Techniques

What is a Type II error in a trading-rule test?

Answer

A Type II error is failing to reject a false null hypothesis, which is a missed real effect. Its probability is beta and statistical power is 1 minus beta.

#247 · Advanced Techniques

How does lowering alpha affect Type I and Type II errors?

Answer

A lower alpha makes the rejection region harder to enter, so Type I errors become less likely and Type II errors usually become more likely. A larger sample can improve power without raising Type I risk.

#248 · Advanced Techniques

Why does sample size affect the ability to detect an edge?

Answer

A larger independent sample reduces the standard error of the sample mean. The same estimated effect becomes easier to distinguish from sampling variation.

#249 · Advanced Techniques

What is the first step in a complete hypothesis-testing sequence?

Answer

Define the market question and the decision that the result will affect. This keeps later choices tied to a useful research purpose.

#250 · Advanced Techniques

Why must hypotheses and test specifications be set before examining outcomes?

Answer

Advance specification prevents the analyst from choosing the most favorable rule or direction after seeing the data. It limits hindsight and unplanned multiple testing.

#251 · Advanced Techniques

What is look-ahead bias?

Answer

It occurs when a historical decision uses information that was unavailable at that time. Revised data, later index membership, or a rule selected from the full sample can create it.

#252 · Advanced Techniques

Why are overlapping forward-return windows a sampling problem?

Answer

Consecutive windows share most of their price data, so their observations are not independent. The nominal sample size overstates the amount of independent evidence.

#253 · Advanced Techniques

What makes a result genuinely out of sample?

Answer

The data were not used to create, select, or tune the rule. Reusing the period after each revision destroys its independence.

#254 · Advanced Techniques

How does the arithmetic mean differ from the geometric mean return?

Answer

The arithmetic mean is the simple average of returns. The geometric mean compounds the growth factors and gives the constant period return that links starting and ending wealth.

#255 · Advanced Techniques

Calculate the geometric mean of returns of 10 percent and negative 5 percent.

Answer

Multiply 1.10 by 0.95 to get 1.045, then take the square root and subtract 1. The result is about 2.225 percent, compared with an arithmetic mean of 2.5 percent.

#256 · Advanced Techniques

What do sample variance and standard deviation measure?

Answer

Variance measures the average squared deviation of returns from their mean, using n minus 1 in a sample. Standard deviation is its square root and is expressed in return units.

#257 · Advanced Techniques

Calculate the z-score for a return of 1.20 percent when the mean is 0.40 percent and the standard deviation is 0.50 percent.

Answer

Subtract the mean from the return, then divide by the standard deviation. The calculation is 0.80 divided by 0.50, so z equals 1.60.

#258 · Advanced Techniques

Why should returns be checked for skew, fat tails, and volatility clustering?

Answer

These features can make a normal-theory test or a single standard deviation misleading. Fat tails, extreme observations, and changing dispersion can require a robust method or a narrower conclusion.

#259 · Advanced Techniques

What does a correlation coefficient measure?

Answer

It measures the direction and strength of a linear association between two variables. It ranges from negative 1 to positive 1 and has no units.

#260 · Advanced Techniques

How is a regression slope interpreted?

Answer

The slope estimates the change in the dependent variable associated with a one-unit change in the explanatory variable. Its units come from the two variables.

#261 · Advanced Techniques

What does an R squared of 0.06 mean?

Answer

The regression explains about 6 percent of the observed variation in the dependent variable within the sample. Most of the variation remains unexplained even if the slope is statistically significant.

#262 · Advanced Techniques

What is a regression residual and why is it inspected?

Answer

A residual is actual y minus fitted y. Residual patterns can reveal curvature, changing error variance, influential points, or serial correlation that the slope alone does not show.

#263 · Advanced Techniques

Why does a predictive regression not establish causation?

Answer

A third factor or a market regime can drive both variables, and the relation can reflect selection or coincidence. Causation requires a credible mechanism and evidence that separates it from alternatives.

#264 · Advanced Techniques

When is a one-sample t-test commonly used?

Answer

It is used to test whether a mean return or mean difference differs from a target when the population standard deviation is unknown. Its interpretation depends on sample size and model assumptions.

#265 · Advanced Techniques

When is a paired test appropriate?

Answer

It is appropriate when observations are matched and share a source of variation, such as the same security and dates under two rules. The test analyzes the paired differences.

#266 · Advanced Techniques

What must be checked before using a two-sample test?

Answer

The groups should match the research design, and their independence and variance structure should be assessed. Separate groups are handled differently from matched observations.

#267 · Advanced Techniques

What is the difference between statistical significance and economic significance?

Answer

Statistical significance concerns whether a result is hard to explain by the null model. Economic significance concerns whether the estimated effect is large enough after costs, turnover, capacity, and risk.

#268 · Advanced Techniques

Why does testing many rules create a false-discovery risk?

Answer

Repeated tests increase the chance that chance variation produces an impressive p-value. The analyst must disclose the search and adjust or validate the final result independently.

#269 · Advanced Techniques

How is a one-sided hypothesis test framed for positive post-signal returns?

Answer

State the null as mean return less than or equal to zero and the alternative as mean return greater than zero. Calculate the standard error, test statistic, critical value, and p-value using that planned direction.

#270 · Advanced Techniques

Why is statistical significance not proof that a trading rule works in the future?

Answer

It describes evidence in one sample under stated assumptions. Implementation costs, changing market conditions, multiple testing, and limited out-of-sample data can remove the apparent edge.

#271 · Advanced Techniques

What does VIX measure?

Answer

VIX measures the S&P 500 expected 30 day volatility implied by SPX option prices. It is annualized and expressed in percentage points.

#272 · Advanced Techniques

What does a VIX value of 20 mean?

Answer

It represents about 20 percent annualized expected volatility. It does not mean that the S&P 500 will move 20 percent over the next month.

#273 · Advanced Techniques

How is the approximate 30 day expected move found from VIX?

Answer

Divide VIX by the square root of 12. A VIX of 20 gives about 5.77 percent for one standard deviation.

#274 · Advanced Techniques

What is the approximate 30 day move when VIX is 28?

Answer

Divide 28 by the square root of 12 to get about 8.08 percent. On an index at 4,200, one standard deviation is about 339 points.

#275 · Advanced Techniques

How is the approximate one day expected move found from VIX?

Answer

Divide VIX by the square root of 252 trading days. A VIX of 24 gives about 1.51 percent for one day.

#276 · Advanced Techniques

What does one standard deviation mean in an expected move calculation?

Answer

It defines a range that contains about 68 percent of outcomes under a simple normal model. Actual market returns have fat tails, so the range is approximate.

#277 · Advanced Techniques

Does a close beyond the VIX expected move prove that VIX was wrong?

Answer

No. The expected move is a probabilistic planning range, not a price ceiling. Outcomes can legitimately fall beyond one standard deviation.

#278 · Advanced Techniques

Does VIX predict the direction of the S&P 500?

Answer

No. VIX estimates the size of expected movement, not whether price will rise or fall.

#279 · Advanced Techniques

What is the main difference between implied and realized volatility?

Answer

Implied volatility comes from option prices and looks forward. Realized volatility comes from past returns and looks backward.

#280 · Advanced Techniques

Why can VIX rise while the S&P 500 moves only a little?

Answer

Demand for options or protection can increase the prices paid for implied volatility. VIX follows those prices before realized movement appears.

#281 · Advanced Techniques

What is the volatility risk premium?

Answer

It is implied volatility minus subsequently realized volatility. A positive value means option prices implied more movement than occurred.

#282 · Advanced Techniques

What does a VRP of plus 6 mean when implied volatility is 21?

Answer

The implied volatility of 21 is six percentage points above realized volatility. Realized volatility is therefore 15 in this example.

#283 · Advanced Techniques

How can a trader attempt to capture the volatility risk premium?

Answer

The trader can sell volatility through option structures such as a short straddle, short strangle, or defined risk credit spread. Each structure has different margin and loss limits.

#284 · Advanced Techniques

Why is selling volatility for the VRP not a free arbitrage?

Answer

Option sellers accept jump, gap, liquidity, and convexity risk. A single severe event can erase many small premiums earned in calm markets.

#285 · Advanced Techniques

What is one difference between a short straddle and a short strangle?

Answer

A short straddle sells a call and a put near the same strike. A short strangle uses out of the money options and usually receives less premium with wider breakeven levels.

#286 · Advanced Techniques

Why can a positive VRP still leave an option seller with a loss?

Answer

The seller can lose from strike placement, path dependent squared returns, transaction costs, margin, and a tail move. Realized variance is not the same as every seller's profit.

#287 · Advanced Techniques

What is contango in VIX futures?

Answer

Contango is an upward sloping term structure in which later VIX futures trade above nearer futures. It is common in calmer markets.

#288 · Advanced Techniques

What is backwardation in VIX futures?

Answer

Backwardation is a downward sloping term structure in which nearer futures trade above later futures. It often appears during stress and strong demand for immediate protection.

#289 · Advanced Techniques

How do you calculate the percentage spread between two VIX futures?

Answer

Divide the later contract price by the front contract price, then subtract one. A rise from 18 to 20 is about 11.1 percent.

#290 · Advanced Techniques

Why can a long position in later VIX futures lose through roll?

Answer

The later contract can decline toward a lower front contract as time passes in contango. This negative roll can create a loss even when VIX changes little.

#291 · Advanced Techniques

Why is a shift from VIX futures contango to backwardation useful as a warning?

Answer

It shows that near term protection and volatility demand are rising relative to later periods. The analyst should confirm it with spreads and other market evidence.

#292 · Advanced Techniques

Does VIX backwardation guarantee a further S&P 500 decline?

Answer

No. It signals a change in near term volatility expectations and hedging demand. It is evidence for risk review, not a guaranteed direction or a certain bottom.

#293 · Advanced Techniques

What are two common ways to manage downside risk in a long S&P 500 portfolio?

Answer

The investor can buy S&P 500 puts or buy VIX calls. The two instruments have different payoff patterns and different types of basis risk.

#294 · Advanced Techniques

How does an S&P 500 put hedge a portfolio?

Answer

A put gains value when the index falls below its strike. It provides a direct floor for an index linked portfolio, subject to cost, expiry, and basis risk.

#295 · Advanced Techniques

Why is a VIX call not a guaranteed substitute for an S&P 500 put?

Answer

A VIX call pays from the behavior of VIX, not from the exact size of an S&P 500 decline. VIX may not rise enough during a moderate price fall to offset portfolio losses.

#296 · Advanced Techniques

Why can a VIX call outperform a put during a panic?

Answer

VIX can spike sharply when hedging demand surges, and a VIX call can have a highly convex payoff. The result still depends on strike, expiry, settlement, and the cost paid.

#297 · Advanced Techniques

What is the key objective of a risk management hedge?

Answer

The objective is to cap a loss the investor is unwilling to tolerate within an acceptable cost and horizon. Maximizing gains in one crisis scenario is not the standard.

#298 · Advanced Techniques

What does Average True Range measure?

Answer

ATR measures the average absolute size of price movement over a selected window, commonly 14 periods. It is used for stop distance and position sizing, not direction.

#299 · Advanced Techniques

How much risk does a stop 2.5 ATR from entry represent when ATR is 20?

Answer

The stop distance is 50 price units. If the account risk budget is $1,000, the position size is at most 20 shares before other costs.

#300 · Advanced Techniques

How is Bollinger BandWidth calculated?

Answer

Divide upper band minus lower band by the middle band. Bands at 110 and 90 around a middle of 100 produce BandWidth of 20 percent.

#301 · Advanced Techniques

What does a Bollinger squeeze show and not show?

Answer

It shows that recent dispersion has contracted. It does not show the direction of the next expansion.

#302 · Advanced Techniques

How is a daily return volatility estimate commonly annualized?

Answer

Multiply the daily standard deviation by the square root of 252. A daily return volatility of 1 percent annualizes to about 15.9 percent under this shortcut.

#303 · Advanced Techniques

How can rising volatility change a trade plan without changing the entry signal?

Answer

Rising volatility can justify smaller position size, a wider stop, or less leverage. The same setup carries more dollar risk when ordinary price movement expands.

#304 · Advanced Techniques

What are the five common trading model families in this chapter?

Answer

They are trend, mean reversion, momentum or relative strength, breakout, and pattern or statistical classification models. Each family makes a different assumption about how market behavior persists and has a different main failure mode.

#305 · Advanced Techniques

What does a trend model assume?

Answer

It assumes that directional price movement persists long enough to produce gains that exceed repeated whipsaw losses. Trend models often perform poorly when the market remains rangebound.

#306 · Advanced Techniques

What is the main risk in a mean-reversion model?

Answer

Price can keep moving away from the reference level instead of returning to it. The model needs a loss limit because frequent small wins can be offset by one large loss during a persistent trend.

#307 · Advanced Techniques

What assumption supports a relative-strength ranking model?

Answer

It assumes that recent leadership persists over the chosen horizon. Crowding, rotation, and high turnover can weaken the result.

#308 · Advanced Techniques

When does a breakout model usually make sense?

Answer

It looks for a confirmed move out of compressed price action or a defined range. The method assumes that the range or compression contains useful information about the next directional move.

#309 · Advanced Techniques

Why is subjective pattern labeling dangerous in a model test?

Answer

An analyst can recognize a pattern after knowing the outcome even if it was not clear at the decision time. A reproducible rule must define every input and decision condition in advance.

#310 · Advanced Techniques

What is the purpose of a go or no-go metric?

Answer

It sets a decision gate for whether a system receives capital, remains under review, or is removed. The threshold should be chosen before testing so results do not rewrite the standard.

#311 · Advanced Techniques

Which ratio compares gross winnings with gross losses?

Answer

Divide gross winning dollars by gross losing dollars. A profit factor of 1.50 means the system won $1.50 for every $1.00 lost before any omitted costs are considered.

#312 · Advanced Techniques

How is expected value per trade calculated?

Answer

Subtract total losses from total wins, then divide by the number of trades. Equivalently, use win rate times average win minus loss rate times average loss.

#313 · Advanced Techniques

Why is a high CAGR by itself weak evidence for implementation?

Answer

It says nothing about the path taken to earn the return. Maximum drawdown, volatility, sample length, costs, and out-of-sample behavior can make the result unsuitable.

#314 · Advanced Techniques

What different risks do the Sharpe and Sortino ratios emphasize?

Answer

Sharpe uses total return volatility in the denominator. Sortino uses downside volatility, so it focuses more directly on harmful moves.

#315 · Advanced Techniques

What does the Calmar ratio compare?

Answer

It commonly compares compound return with maximum drawdown. A higher value means more return per unit of worst historical drawdown.

#316 · Advanced Techniques

What are the four practical ways to generate a trading signal?

Answer

They are level crossing, crossover, threshold crossing, and multi-condition state recognition. Each must define its data timing and confirmation rule.

#317 · Advanced Techniques

What is the difference between a level crossing and a crossover?

Answer

A level crossing compares price or an indicator with a fixed or evolving boundary. A crossover compares two series, such as a fast and slow moving average.

#318 · Advanced Techniques

How should a model handle a signal that uses the closing price?

Answer

The signal becomes known only after the close. The test should trade at the next available price or explicitly model a close fill while recognizing that the fill cannot occur before the information is known.

#319 · Advanced Techniques

What is a trigger rule?

Answer

A trigger is the event that causes action, such as a confirmed breakout or moving-average crossover. It answers what makes the model act now.

#320 · Advanced Techniques

What is a filter rule?

Answer

A filter permits or blocks a trigger under stated conditions. It does not create the trade by itself and usually reduces the number of signals.

#321 · Advanced Techniques

What is a value rule?

Answer

A value rule acts when price is far from a reference measure of worth. It commonly supports mean-reversion decisions.

#322 · Advanced Techniques

What does a signal test measure?

Answer

It measures the forward return after an entry condition and compares it with a relevant baseline. It does not include the full strategy's stops, sizing, or portfolio constraints.

#323 · Advanced Techniques

What does a full backtest add beyond a signal test?

Answer

It simulates entries, exits, stops, position sizing, overlapping signals, available capital, costs, and realistic fills. It tests the whole strategy rather than the isolated signal.

#324 · Advanced Techniques

Why should a signal test usually come before a full backtest?

Answer

A profitable backtest mixes entry, exit, and sizing decisions, so a strong exit can hide a weak signal. Testing the signal first isolates whether the entry contains information.

#325 · Advanced Techniques

Why are 300 clustered signals not 300 independent observations?

Answer

Overlapping holding periods cause many signals to share the same subsequent price moves. The effective sample is smaller, so ordinary significance calculations can overstate confidence.

#326 · Advanced Techniques

Name the seven stops in the layered risk process.

Answer

They are the initial stop, trailing stop, volatility stop, time stop, signal stop, portfolio stop, and catastrophic stop.

#327 · Advanced Techniques

What is the purpose of an initial stop?

Answer

It defines the planned maximum loss when the trade opens. It also makes risk per share known so position size can be calculated.

#328 · Advanced Techniques

How does a trailing stop differ from an initial stop?

Answer

A trailing stop moves with a favorable trend and generally does not move backward. It protects accumulated gains instead of setting only the opening risk.

#329 · Advanced Techniques

How is a 2.5 ATR volatility stop calculated from a $100 entry and $2.00 ATR?

Answer

Multiply 2.5 by $2.00 to get a $5.00 stop distance. Subtract $5.00 from $100 to place the stop at $95.

#330 · Advanced Techniques

What is the purpose of a time stop?

Answer

It exits when the expected move has not developed within a defined period. This prevents a dead trade from tying up capital and risk indefinitely.

#331 · Advanced Techniques

When should a signal stop be used?

Answer

Use it when the original reason for the trade has failed even if the price has not reached the money stop. It exits because the thesis is no longer valid.

#332 · Advanced Techniques

What does a portfolio stop control?

Answer

It controls aggregate exposure such as total open risk, correlated positions, gross exposure, or new entries after a portfolio drawdown. It addresses risks that separate position stops cannot see.

#333 · Advanced Techniques

What is the role of a catastrophic stop?

Answer

It is an emergency account-level rule for gaps, broken data, platform failure, or losses beyond the normal model envelope. It preserves the process for review after severe disruption.

#334 · Advanced Techniques

What does optimization do, and what risk does it create?

Answer

Optimization searches parameter values for strong historical results. Searching too many values can select random noise and create overfitting.

#335 · Advanced Techniques

What is overfitting?

Answer

Overfitting is tuning a rule so closely to one sample that it captures accidental details instead of a repeatable relationship. The model then tends to disappoint on new data.

#336 · Advanced Techniques

What does a robust parameter region look like?

Answer

Nearby values also produce acceptable results, forming a plateau rather than one isolated peak. Robustness should also survive changes in costs, timing, universe, and market regime.

#337 · Advanced Techniques

How should out-of-sample and walk-forward testing be used?

Answer

Out-of-sample data must play no part in building or selecting the rule. Walk-forward testing repeatedly chooses parameters on one window and applies them to the next untouched window, creating a sequence of forward tests.

#338 · Advanced Techniques

What does total return include?

Answer

Total return includes price change plus distributions received over the measurement period. It is expressed relative to the starting investment value.

#339 · Advanced Techniques

How is a simple total return calculated?

Answer

Subtract the beginning value from the ending value plus distributions, then divide by the beginning value. Include costs when measuring the actual investor result.

#340 · Advanced Techniques

How does the arithmetic average return differ from the geometric average return?

Answer

The arithmetic average adds periodic returns and divides by their count. The geometric average compounds the growth factors and reports the constant rate that links beginning and ending wealth.

#341 · Advanced Techniques

What causes volatility drag?

Answer

Volatility drag arises because percentage returns compound multiplicatively, so varying returns generally produce a geometric average below the arithmetic average. A loss also requires a larger percentage recovery than the size of the loss.

#342 · Advanced Techniques

What is the formula for CAGR?

Answer

CAGR equals (ending value / beginning value)^(1 / years) - 1. It converts compound growth over a period into a constant annual growth rate.

#343 · Advanced Techniques

What does CAGR hide about a portfolio path?

Answer

CAGR smooths all interim gains, losses, and drawdowns into one annual rate. Two portfolios can have the same CAGR while one experienced much greater capital damage.

#344 · Advanced Techniques

How is a short holding period return annualized simply?

Answer

Multiply the period return by the number of such periods in a year. A 3% return over 90 days scales to 3% × 365 / 90 = 12.17% under simple annualization.

#345 · Advanced Techniques

Why should a stated annualization convention be checked?

Answer

Simple and compound annualization can produce different results. Comparing a simple annualized return with a CAGR without recognizing the method can create a false ranking.

#346 · Advanced Techniques

How is annualized volatility estimated from daily returns?

Answer

Multiply the daily return standard deviation by the square root of the number of trading periods in a year. A 1.2% daily standard deviation scales to about 19.05% with 252 periods.

#347 · Advanced Techniques

What assumptions support volatility scaling?

Answer

The square root scaling rule assumes returns are reasonably independent and volatility is stable. Autocorrelation, volatility clustering, and fat tails can make the estimate incomplete.

#348 · Advanced Techniques

What is maximum drawdown?

Answer

Maximum drawdown is the largest peak to trough percentage decline in a wealth series. It measures an experienced path loss rather than volatility around an average return.

#349 · Advanced Techniques

Why should drawdown duration be reported with maximum drawdown?

Answer

Maximum drawdown gives the depth of the worst decline, while duration gives the time needed to recover or to move from peak to trough. A deep drawdown and a long recovery pose different risks.

#350 · Advanced Techniques

What does downside deviation measure?

Answer

Downside deviation measures dispersion below a chosen target return while assigning no downside penalty to returns at or above the target. The target and denominator convention must be stated.

#351 · Advanced Techniques

How is profit factor calculated?

Answer

Divide gross profits from winning trades by the absolute value of gross losses from losing trades. A profit factor above 1.00 means gross profits exceeded gross losses.

#352 · Advanced Techniques

Why can profit factor mislead?

Answer

It does not show required capital, trade duration, costs, or dependence on one exceptional winner. The distribution of profits and losses must be inspected with it.

#353 · Advanced Techniques

How is expected value per trade calculated from win rate and average outcomes?

Answer

Multiply the win rate by average win, then subtract the loss rate multiplied by average loss. The result is the estimated average profit or loss per trade before costs.

#354 · Advanced Techniques

What is payoff ratio?

Answer

Payoff ratio is average winning trade divided by average losing trade. A high payoff ratio can make a lower win rate profitable.

#355 · Advanced Techniques

What is break even win rate for a given payoff ratio?

Answer

Break even win rate equals average loss divided by the sum of average win and average loss. Actual costs raise the win rate required for positive expectancy.

#356 · Advanced Techniques

How is average trade calculated?

Answer

Divide total net profit by the number of trades. Comparisons are useful only when capital, exposure, holding periods, and sample conditions are reasonably similar.

#357 · Advanced Techniques

What is maximum favorable excursion?

Answer

Maximum favorable excursion is the greatest unrealized gain reached after entry in the trade's intended direction. It shows how much favorable movement was available before exit.

#358 · Advanced Techniques

What is maximum adverse excursion?

Answer

Maximum adverse excursion is the greatest unrealized loss reached while a trade is open. It measures how far price moved against the position before exit.

#359 · Advanced Techniques

How is MAE calculated for a long trade?

Answer

Subtract the lowest price reached while the trade is open from the entry price. A long entered at 50 that falls to 47.20 has MAE of 2.80 per share.

#360 · Advanced Techniques

What can MAE reveal about eventual winners?

Answer

Large adverse movement among trades that eventually win can indicate stops are too tight or entries are poorly timed. It should be studied across a sample before changing rules.

#361 · Advanced Techniques

What can repeated surrender of MFE reveal?

Answer

Large favorable excursion that is routinely given back can point to weak exit rules or slow trade management. A single trade cannot establish a reliable pattern.

#362 · Advanced Techniques

How is the Sharpe ratio calculated?

Answer

Divide return minus the risk free rate by return standard deviation. It measures excess return per unit of total return volatility.

#363 · Advanced Techniques

Why must Sharpe ratio inputs use the same frequency?

Answer

Mixing a monthly volatility with an annual risk free rate or return creates a malformed ratio. Return, risk free rate, and volatility must use matching period and annualization conventions.

#364 · Advanced Techniques

How is the Sortino ratio calculated?

Answer

Divide return minus the target return by downside deviation. The result focuses on excess return relative to harmful variation below the stated target.

#365 · Advanced Techniques

How is the Calmar ratio typically calculated?

Answer

Divide annualized return by the absolute value of maximum drawdown. It relates reward to the worst historical peak to trough decline, so sample length matters.

#366 · Advanced Techniques

How is the information ratio calculated?

Answer

Divide active return by tracking error. Active return is portfolio return minus benchmark return, and tracking error is the standard deviation of active returns.

#367 · Advanced Techniques

Which performance ratio fits a benchmark relative mandate?

Answer

The information ratio fits because it measures active return per unit of benchmark relative variability. Sharpe ratio measures absolute return per unit of total volatility.

#368 · Advanced Techniques

Which measures deserve emphasis for a trend following strategy?

Answer

Inspect CAGR, maximum drawdown, drawdown duration, Calmar ratio, profit factor, and dependence on the largest winners. Such strategies often combine low win rates with large gains.

#369 · Advanced Techniques

Which risks deserve extra attention for a mean reversion strategy?

Answer

Inspect Sharpe and Sortino ratios along with tail losses, maximum drawdown, holding period, and recovery time. A high win rate can coexist with infrequent losses that erase many gains.

#370 · Advanced Techniques

Why should performance be judged on net and out of sample results?

Answer

Net results include implementation costs, while out of sample results test whether a rule survives data not used to optimize it. A strong in sample ratio can be the product of curve fitting.

#371 · Advanced Techniques

What does a rising relative strength ratio line show?

Answer

It shows that the asset gained more or lost less than the benchmark over the measured period. It does not show that the asset's absolute price increased.

#372 · Advanced Techniques

How is a basic relative strength ratio calculated?

Answer

Divide the asset price by the benchmark price after putting both series on a consistent scale. The direction and slope of the ratio matter more than the initial numerical level.

#373 · Advanced Techniques

How is relative return calculated from two absolute returns?

Answer

Divide one plus the asset return by one plus the benchmark return, then subtract one. If the asset gains 5% and the benchmark gains 2%, relative return is 1.05 / 1.02 - 1 = 2.94%.

#374 · Advanced Techniques

Can an asset have positive relative strength while its price falls?

Answer

Yes. A stock that falls 5% while its benchmark falls 8% loses money in absolute terms but outperforms the benchmark. The analyst must describe both dimensions.

#375 · Advanced Techniques

Why can a linear relative strength chart distort a long comparison?

Answer

Equal absolute moves receive equal vertical space even when percentage changes differ. A logarithmic scale shows equal percentage changes as equal distances and can alter the visual reading of trend and turning points.

#376 · Advanced Techniques

What is the first major issue in relative strength analysis?

Answer

The denominator or benchmark must fit the decision being made. A stock can lead its sector, lag a broad index, or show a different result after currency adjustment.

#377 · Advanced Techniques

What must be consistent across the numerator and denominator of a price ratio?

Answer

Both series should use the same currency, date range, price adjustment convention, and data timing. Split or dividend adjustments should also be handled consistently.

#378 · Advanced Techniques

How does a relative strength ratio differ from the Relative Strength Index?

Answer

The ratio compares two price series and has no fixed upper or lower bound. RSI is a bounded momentum oscillator calculated from one price series.

#379 · Advanced Techniques

Can an asset have a high RSI and a falling relative strength ratio?

Answer

Yes. The asset can have strong one-series momentum while still underperforming its benchmark. The labels measure different relationships.

#380 · Advanced Techniques

What does the centered value of 100 mean on a JdK RS chart?

Answer

It represents a neutral reading near the rolling norm used by the chart. Readings above 100 are above that norm and readings below 100 are below it.

#381 · Advanced Techniques

What does JdK RS-Ratio measure?

Answer

It measures the normalized level of the relative strength ratio over a rolling window. It shows where the current ratio sits relative to its recent average and dispersion.

#382 · Advanced Techniques

What does JdK RS-Momentum measure?

Answer

It measures the normalized speed of change in the JdK RS-Ratio rather than the speed of change in absolute price. A reading above 100 indicates rising relative momentum.

#383 · Advanced Techniques

How can an asset plot at 115 for RS-Ratio when its raw ratio is 1.25?

Answer

The platform first standardizes the ratio against its rolling mean and standard deviation, then maps the resulting score around 100. Exact scaling can differ by platform.

#384 · Advanced Techniques

Why can RS-Momentum turn below 100 before RS-Ratio falls below 100?

Answer

Relative momentum measures the change in relative level. The ratio can remain above its norm while the pace of its rise slows or turns negative.

#385 · Advanced Techniques

What are the axes of a Relative Rotation Graph?

Answer

JdK RS-Ratio is on the horizontal axis and JdK RS-Momentum is on the vertical axis. Each axis is centered at 100.

#386 · Advanced Techniques

Which RRG values define the Leading quadrant?

Answer

Both JdK RS-Ratio and JdK RS-Momentum are above 100. The asset is relatively strong and its relative strength is still improving.

#387 · Advanced Techniques

Which RRG values define the Weakening quadrant?

Answer

JdK RS-Ratio is above 100 while JdK RS-Momentum is below 100. The asset remains relatively strong but its relative momentum has declined.

#388 · Advanced Techniques

Which RRG values define the Lagging quadrant?

Answer

Both JdK RS-Ratio and JdK RS-Momentum are below 100. The asset is relatively weak and relative momentum is negative.

#389 · Advanced Techniques

Which RRG values define the Improving quadrant?

Answer

JdK RS-Ratio is below 100 while JdK RS-Momentum is above 100. Relative level is still weak, but relative momentum has turned up.

#390 · Advanced Techniques

What is the ideal clockwise RRG rotation?

Answer

The sequence is Improving, Leading, Weakening, Lagging, then back toward Improving. The path can be irregular because market evidence changes at different speeds.

#391 · Advanced Techniques

How is a point at (97, 105) classified on an RRG?

Answer

It is in Improving because RS-Ratio is below 100 and RS-Momentum is above 100. The asset is still below its recent relative norm but is gaining relative strength.

#392 · Advanced Techniques

Where does the benchmark appear on an RRG?

Answer

The benchmark is centered at 100 on each axis because its relative ratio with itself is constant. Every asset's coordinates are measured against that benchmark.

#393 · Advanced Techniques

What happens to RRG readings when the benchmark changes?

Answer

The coordinates, quadrants, and rotation paths can change because the denominator has changed. Each map must be read against one consistent benchmark.

#394 · Advanced Techniques

What information does an RRG trail add beyond a single quadrant?

Answer

The trail shows direction, speed, persistence, and whether the asset is entering or leaving a state. A long sustained path is stronger evidence than one point near the center.

#395 · Advanced Techniques

Why is an RRG quadrant not a direct buy or sell signal?

Answer

Quadrants describe relative level and relative momentum. They do not establish absolute price trend, valuation, liquidity, or portfolio risk.

#396 · Advanced Techniques

What is the role of credit spreads in cross-asset analysis?

Answer

They show the extra yield investors demand for corporate credit risk. Broad widening can indicate tighter financing conditions or weaker risk appetite.

#397 · Advanced Techniques

Why should foreign returns be compared in one currency?

Answer

Currency moves can reverse the ranking seen in local prices. A foreign index can rise locally and lose money for a domestic investor after translation.

#398 · Advanced Techniques

How can a stronger dollar affect cross-asset evidence?

Answer

It can create translation pressure for exporters and dollar-priced commodities, while affecting commodity-linked currencies differently. The signal depends on the assets and mechanism.

#399 · Advanced Techniques

Why should the reason for a rate move be considered?

Answer

The same yield increase can reflect stronger growth, higher inflation expectations, or a changing term premium. Those causes can affect equities and credit differently.

#400 · Advanced Techniques

Which three evidence types are combined in a relative asset weighting model?

Answer

The model commonly combines relative strength, absolute trend, and price momentum. Each answers a different part of the leadership question.

#401 · Advanced Techniques

How is a weighted composite score calculated?

Answer

Multiply each standardized or percentile input by its assigned weight, then add the contributions. With scores 80, 70, and 90 and weights 40%, 35%, and 25%, the composite is 79.0.

#402 · Advanced Techniques

Why can a relative strength ranking alone be unsuitable for a long-only portfolio?

Answer

An asset can outperform a falling benchmark while still losing capital. Absolute trend and portfolio risk filters must be considered separately.

#403 · Advanced Techniques

What confirms a sector leadership signal most strongly?

Answer

A rising relative strength line combined with a healthy absolute price trend and consistent evidence from price, momentum, or market internals provides stronger confirmation.

#404 · Advanced Techniques

What is account risk per trade?

Answer

Account risk is the maximum planned dollar loss allowed on one trade. It is usually set as a percentage of account equity before position sizing begins.

#405 · Advanced Techniques

How is stop distance measured for a long trade?

Answer

Subtract the stop price from the entry price. If entry is 80 and the stop is 76.50, stop distance is 3.50 per share.

#406 · Advanced Techniques

What is the position sizing formula for a long trade?

Answer

For shares, divide planned account risk by stop distance. For contracts, divide by price distance multiplied by the contract multiplier or point value. Round down to whole units.

#407 · Advanced Techniques

Why is position value not the same as trade risk?

Answer

Position value is the full market value controlled, while planned trade risk is only the loss from entry to the stop. A large position with a tight stop can have limited planned risk.

#408 · Advanced Techniques

How is position size calculated for a short sale?

Answer

Divide the risk budget by the amount the stop is above the short entry. Entering at 120 with a 126.75 stop creates 6.75 of risk per share.

#409 · Advanced Techniques

What should determine the initial stop distance?

Answer

The price where the setup is genuinely invalidated should determine the stop. The stop should also allow the normal noise seen in similar trades.

#410 · Advanced Techniques

How is an ATR stop distance set?

Answer

Multiply ATR by a chosen multiple that reflects the trade timeframe and normal noise. A 2.5 ATR stop with 2.00 ATR places the stop 5.00 away before entry slippage.

#411 · Advanced Techniques

What is a technical stop?

Answer

A technical stop is placed beyond a chart level whose failure rejects the trade. Examples include a prior swing low, pattern neckline, or failed support.

#412 · Advanced Techniques

How does a trailing stop work?

Answer

A trailing stop moves in the favorable direction and does not move against the position. It can follow higher lows or trail a chosen ATR distance behind price.

#413 · Advanced Techniques

What is a time stop?

Answer

A time stop exits a trade when the expected price move has not occurred within a specified number of bars. It prevents a stalled thesis from using capital indefinitely.

#414 · Advanced Techniques

Why is a stop not a guaranteed maximum loss?

Answer

Price can gap through the stop or fill with slippage after activation. Position sizing must consider a loss larger than the distance to the stated stop.

#415 · Advanced Techniques

How is reward to risk calculated for a long trade?

Answer

Divide target minus entry by entry minus stop. An 8.00 target gain with 4.00 of risk gives reward to risk of 2.0 to 1.

#416 · Advanced Techniques

What makes a price target technically defensible?

Answer

A defensible target comes from chart evidence such as a measured move, prior supply zone, pattern projection, or confluence level. Its reward must reflect a realistic obstacle.

#417 · Advanced Techniques

What is the break even win rate for a fixed reward to risk ratio?

Answer

It is 1 divided by 1 plus the reward to risk ratio. A 3 to 1 ratio has a 25% break even win rate before transaction costs.

#418 · Advanced Techniques

How is trade expectancy calculated?

Answer

Multiply win rate by average win and subtract loss rate multiplied by average loss. A 45% win rate with 1,200 average win and 500 average loss yields 265 per trade.

#419 · Advanced Techniques

Why can a high win rate still lose money?

Answer

One or more losses can be much larger than the frequent small gains. Expectancy and loss size matter more than the winning percentage by itself.

#420 · Advanced Techniques

How does leverage affect an account loss?

Answer

Leverage multiplies the percentage gain or loss in the underlying asset. Three times exposure to a 10% decline produces about a 30% account loss before costs.

#421 · Advanced Techniques

What percentage gain recovers a 20% loss?

Answer

A 20% loss leaves 80% of the original wealth. Reaching the prior high requires 20 / 80 = 25%, because the gain starts from a smaller base.

#422 · Advanced Techniques

What percentage gain recovers a 40% loss?

Answer

A 40% loss leaves 60% of the original wealth. It requires a 66.7% gain because 40 / 60 = 0.667.

#423 · Advanced Techniques

Why is drawdown recovery asymmetric?

Answer

A loss reduces the capital base on which the recovery gain is calculated. Larger losses therefore require progressively larger percentage gains to regain the prior peak.

#424 · Advanced Techniques

How does leverage affect survival probability?

Answer

Leverage enlarges losses and can trigger margin calls or forced sales during ordinary volatility. Survival depends on the worst sequence of losses, gaps, and financing costs.

#425 · Advanced Techniques

What is portfolio heat?

Answer

Portfolio heat is the sum of planned losses across open positions if their current stops are reached. It is usually compared with total account equity.

#426 · Advanced Techniques

How is portfolio heat calculated in percentage terms?

Answer

Divide total planned dollar loss by account equity. Planned losses of 4,000 on a 100,000 account equal 4.0% heat.

#427 · Advanced Techniques

Why can portfolio heat understate simultaneous risk?

Answer

Heated positions may share a sector, factor, currency, or liquidity shock. Correlated stops can gap at the same time instead of occurring independently.

#428 · Advanced Techniques

What is risk contribution?

Answer

Risk contribution is the amount of portfolio risk supplied by a position or group of positions. A small allocation to a volatile asset can contribute more risk than a larger allocation to a quiet asset.

#429 · Advanced Techniques

Why does diversification fail in some market shocks?

Answer

Correlations can rise when stressed positions are sold together. Assets that seemed independent in a calm sample may become one common source of loss.

#430 · Advanced Techniques

What is gap risk in a trade plan?

Answer

Gap risk is the chance that the market opens beyond the stop price. The actual exit loss can be much larger than the planned entry to stop distance.

#431 · Advanced Techniques

How should a stress test differ from a normal stop calculation?

Answer

A stress test assumes the stop is skipped or filled far away under a plausible shock. It estimates the account result from gap, correlation, and liquidity pressure.

#432 · Advanced Techniques

Why is liquidity risk part of position sizing?

Answer

A position that is large relative to available volume may move the price during entry or exit. Market impact and wider spreads raise the loss beyond the chart based stop.

#433 · Advanced Techniques

What does model failure mean in risk management?

Answer

Model failure means the rule no longer performs within its tested environment or its assumptions break. Position and drawdown limits should protect the account while the evidence is reassessed.

#434 · Advanced Techniques

How is a volatility targeting exposure multiplier calculated?

Answer

Divide target volatility by estimated asset volatility. A 10% target with 20% estimated volatility gives a 0.50 exposure multiplier.

#435 · Advanced Techniques

How can signal quality affect position size?

Answer

Stronger evidence can receive more of the risk budget than a marginal setup, provided the adjustment follows prewritten rules. Every tier still needs a hard maximum.

#436 · Advanced Techniques

What should a complete pre trade risk plan include?

Answer

It should name entry, invalidation, size, maximum planned loss, target, reward to risk, heat, leverage, stress loss, and the evidence that cancels the trade.

#437 · Application of Technical Analysis

Why should a technical trade plan rank the evidence?

Answer

The plan should identify primary evidence, confirming evidence, conflicts, and the event that falsifies the thesis. Ranking keeps a minor indicator from overriding market structure.

#438 · Application of Technical Analysis

What is usually the primary evidence for a trend-following setup?

Answer

Price structure and the direction of the established trend are primary. Volume and momentum can confirm the trend, while sentiment and volatility check crowding and risk.

#439 · Application of Technical Analysis

What does heavy breakout volume with a close near the session high show?

Answer

It shows demand accepting the higher price during the session. Heavy volume with a weak close instead suggests that supply absorbed the breakout attempt.

#440 · Application of Technical Analysis

Why is broadening breadth useful evidence at an index high?

Answer

It shows that more stocks are participating instead of a small group carrying the index. Narrow leadership makes the new high less reliable.

#441 · Application of Technical Analysis

How should a high RSI be read inside a persistent uptrend?

Answer

It can signal persistent strength rather than an automatic reversal. An overbought reading matters more as a warning when price structure also weakens.

#442 · Application of Technical Analysis

What is the best use of an extreme sentiment reading?

Answer

Use it as a measure of crowded positioning and vulnerability. Price rejection or a failed breakout provides the timing evidence.

#443 · Application of Technical Analysis

How does volatility affect position sizing?

Answer

Higher volatility requires a wider stop for the same technical invalidation or a smaller position for the same dollar risk. Volatility changes the size of normal price swings.

#444 · Application of Technical Analysis

What makes an entry rule objective?

Answer

It states an observable event that must occur before capital is committed, such as a close above resistance with confirming volume. A general bullish opinion is not an entry rule.

#445 · Application of Technical Analysis

How does thesis invalidation differ from a protective stop?

Answer

Invalidation is the market event that makes the original evidence false. The protective stop is an order placed beyond that event with a buffer for noise and execution.

#446 · Application of Technical Analysis

What two jobs should an exit plan perform?

Answer

One exit captures the planned target and another protects profit as the trade develops. Each exit should have a stated technical or time based reason.

#447 · Application of Technical Analysis

Entry is 99, invalidation is 95.50, and the target is 107. What is the reward to risk ratio?

Answer

The reward is 8.00 points and the risk is 3.50 points. The ratio is 8.00 divided by 3.50, or 2.29.

#448 · Application of Technical Analysis

What formula converts account risk into a basic share count?

Answer

Shares equal the account equity times the risk fraction, divided by risk per share. Round down to avoid exceeding the dollar risk budget.

#449 · Application of Technical Analysis

A position is 50 percent of a portfolio and has a 4 percent stop. What is the portfolio loss before slippage?

Answer

The portfolio loss is 50 percent times 4 percent, or 2 percent. Position weight scales the impact of the stop.

#450 · Application of Technical Analysis

With three times equity exposure, what asset decline produces a 10 percent account loss?

Answer

The asset decline is 10 percent divided by 3, or about 3.3 percent. Leverage magnifies the account impact of the price move.

#451 · Application of Technical Analysis

Why can several separate charts still create one concentrated risk?

Answer

They may share the same sector, factor, currency, or liquidity exposure. Aggregate those common risks instead of treating each chart as independent.

#452 · Application of Technical Analysis

Why is a positive backtest insufficient evidence for using a model?

Answer

The test may reflect overfitting, costs, limited capacity, or unusual market conditions. Out of sample results and current regime fit must also be reviewed.

#453 · Application of Technical Analysis

What is an example of a model and decision purpose mismatch?

Answer

A short horizon trend model used for a monthly allocation is poorly aligned with its purpose. The signal horizon should match the decision horizon.

#454 · Application of Technical Analysis

What does out of sample mean in model testing?

Answer

It means testing on data that was not used to build the model or choose its parameters. It provides a cleaner test of whether the rule has general value.

#455 · Application of Technical Analysis

What is parameter robustness?

Answer

Robustness means similar parameter values produce reasonably similar results. A narrow peak of spectacular performance is less credible than a broad stable region.

#456 · Application of Technical Analysis

What is regime fit in model suitability?

Answer

Regime fit asks whether current volatility, trend, liquidity, and correlation conditions resemble the conditions in which the model was tested. A valid historical model can be unsuitable in a different environment.

#457 · Application of Technical Analysis

What actions can follow a model suitability review?

Answer

The analyst can use the model at normal size, use it at reduced size, pause it for more evidence, or reject its output. Rejection is a valid decision.

#458 · Application of Technical Analysis

How does relative strength differ from RSI?

Answer

Relative strength compares an asset with a benchmark, while RSI measures the asset's own momentum. A relative strength line can rise while both the asset and benchmark fall.

#459 · Application of Technical Analysis

Why combine relative strength, trend, and momentum in an asset ranking?

Answer

The three measures answer different questions about leadership, direction, and speed. A blend can separate improving leadership from a rally driven only by the market.

#460 · Application of Technical Analysis

Three assets have volatilities of 20, 25, and 50 percent. What do inverse volatility weights show?

Answer

The inverse values are 5, 4, and 2, which give weights of 45.5, 36.4, and 18.2 percent. The least volatile asset receives the largest raw weight.

#461 · Application of Technical Analysis

Why apply a concentration cap after calculating model weights?

Answer

A cap prevents one high score or low volatility estimate from creating excessive single asset risk. The excess weight must be assigned by a documented rule or held in cash.

#462 · Application of Technical Analysis

When should a ranked asset be removed from a portfolio?

Answer

Common exits include a sustained drop below the eligibility rank, a trend failure, or a confirmed turn in relative strength. Periodic review keeps the portfolio aligned with the model.

#463 · Application of Technical Analysis

VIX is 24. What is the approximate 30 day one standard deviation move?

Answer

The move is 24 divided by the square root of 12, which equals 6.93 percent. This uses the conventional 12 periods per year shortcut.

#464 · Application of Technical Analysis

VIX is 24. What is the approximate one day implied move?

Answer

The move is 24 divided by the square root of 252, which equals about 1.51 percent. The estimate is a range input, not a guarantee.

#465 · Application of Technical Analysis

What is a major limitation of a VIX based price forecast?

Answer

It describes a range under model assumptions and does not reveal direction. The volatility estimate can also change before the forecast horizon ends.

#466 · Application of Technical Analysis

Support is 94, resistance is 106, and ATR is 2.50. How many ATRs is the target move?

Answer

The distance is 12 points and 12 divided by 2.50 equals 4.8 ATRs. The analyst can use that value to judge whether the target is realistic for the holding period.

#467 · Application of Technical Analysis

Why can a stop only 0.4 percent from entry be unreliable when the daily implied move is 1.5 percent?

Answer

Ordinary daily noise is much larger than the stop distance. A wider technical stop requires a smaller position to keep account risk controlled.

#468 · Application of Technical Analysis

What does a Bollinger BandWidth squeeze actually identify?

Answer

It identifies contracted volatility and the possibility of expansion. It does not identify the direction of the eventual break.

#469 · Application of Technical Analysis

How should price structure and a lagging oscillator conflict be resolved?

Answer

Price structure normally carries more weight than a minor lagging oscillator conflict. A break of the main trend changes the thesis more than a temporary momentum wobble.

#470 · Application of Technical Analysis

In what order should integrated evidence be reviewed?

Answer

Review regime and price confirmation first, then participation, positioning, volatility, and model fit. This order prevents sentiment from overriding a failed setup.

#471 · Application of Technical Analysis

Why does an extreme sentiment reading support waiting for price confirmation?

Answer

Crowding shows vulnerability but does not show timing. A failed breakout or reversal close connects the positioning evidence to an actionable signal.

#472 · Application of Technical Analysis

Why must trade process be reviewed separately from profit or loss?

Answer

A good process can lose and a poor process can win by chance. Rewarding a poor process because one outcome was favorable encourages repetition of the error.

#473 · Application of Technical Analysis

What does maximum adverse excursion measure?

Answer

It measures how far a trade moved against entry before it closed. A deep adverse move can reveal an early entry, a tight stop, or excessive size.

#474 · Application of Technical Analysis

What does maximum favorable excursion help review?

Answer

It measures how far a trade moved in the intended direction during its life. Comparing it with the exit can show whether the exit process captured available profit.

#475 · Application of Technical Analysis

How should an analyst choose between two defensible trade plans?

Answer

Choose the plan with the clearer trigger, stronger evidence alignment, better defined invalidation, and more appropriate size. Both plans can be reasonable while one better fits the facts.

#476 · Application of Technical Analysis

Why reduce a breakout model when VIX is above its tested range?

Answer

Higher volatility increases gap and stop out risk while reducing the reliability of the historical breakout pattern. Reduced size keeps the plan inside its tested assumptions.

#477 · Application of Technical Analysis

How should a bearish pattern be managed when put call sentiment is already extremely bearish?

Answer

Treat the pattern as valid only after its price trigger and reduce exposure for squeeze risk. A smaller position respects the conflict between technical weakness and crowded short positioning.

#478 · Application of Technical Analysis

Why can an earnings date make a valid pattern model unsuitable overnight?

Answer

An event gap can jump past the planned stop and create losses larger than the tested model assumes. Postponing the trade keeps the process inside its risk limits.

#479 · Application of Technical Analysis

When should volatility scaling reduce a position even if the dollar stop remains valid?

Answer

Reduce size when the position would contribute too much forecast portfolio volatility or gap risk. A valid stop does not by itself make the exposure suitable.

#480 · Application of Technical Analysis

Why is a confirmed trigger stronger than entering from anticipation?

Answer

The trigger shows that buyers or sellers have accepted the new price area. Anticipation commits risk before the setup has earned confirmation.

#481 · Application of Technical Analysis

A basic size is 333 shares and gap risk requires a 50 percent reduction. What is the adjusted size?

Answer

The adjusted size is 333 times 0.50, which equals 166.5 shares. Round down to 166 shares.

#482 · Application of Technical Analysis

What evidence can justify rejecting a profitable backtest?

Answer

Reject or pause it when out of sample results fail, costs erase the edge, capacity is poor, parameter stability is weak, or current conditions are far outside the test regime. A profitable sample does not prove implementation suitability.

#483 · Application of Technical Analysis

What elements belong in a complete integrated trade plan?

Answer

The plan should state the thesis, entry trigger, exit targets, invalidation event, position size, portfolio exposure, and model suitability. It should also include the evidence that would change the view.

#484 · Application of Technical Analysis

Why can a rising relative strength line occur in a falling market?

Answer

The asset can fall less than its benchmark while still losing price. Relative strength measures comparative performance rather than absolute direction.

#485 · Application of Technical Analysis

How should a plan respond when volatility expands before the entry trigger?

Answer

Reassess the stop distance, target distance, and position size before committing capital. If the revised risk no longer supports the objective, cancel the plan.

#486 · Application of Technical Analysis

What connects the evidence in a strong applied technical answer?

Answer

The answer should connect evidence, trigger, invalidation, size, and model suitability into one consistent decision. A direction alone is not a complete trade plan.

#487 · Ethics

Which professional framework applies to the CMT ethics questions?

Answer

The CMT Association uses the CFA Institute Code of Ethics and Standards of Professional Conduct. The CMT Association states that references to CFA Institute members, candidates, and programs apply to the CMT Association and its members, candidates, and programs.

#488 · Ethics

What six broad commitments appear in the Code of Ethics?

Answer

The Code requires integrity, competence, diligence, respect, and ethical conduct. It also requires client interests and professional integrity to come before personal interests, independent judgment, ethical leadership, support for sound global markets, and continuing competence.

#489 · Ethics

How should a professional act when a law and the Code conflict?

Answer

The professional must comply with the stricter requirement. The professional must also avoid knowingly participating in or assisting a violation and must dissociate from continuing misconduct.

#490 · Ethics

What does the Competence standard require?

Answer

A member or candidate must possess and maintain the competence needed to perform professional responsibilities. An unfamiliar model or product should not be used for clients until the professional has adequate knowledge and support.

#491 · Ethics

When does a gift threaten independence and objectivity?

Answer

A gift is prohibited when it could reasonably be expected to compromise professional judgment. Source, purpose, timing, frequency, and value all matter, so disclosure alone may not fix a lavish issuer-paid benefit.

#492 · Ethics

What must an analyst do after receiving material nonpublic information?

Answer

The analyst must not trade, change a recommendation, or cause others to act on the information. The analyst should involve compliance and encourage the issuer to make prompt public disclosure.

#493 · Ethics

How does mosaic theory permit an analyst to reach a material-looking conclusion?

Answer

The analyst may combine public information, nonmaterial nonpublic observations, and independent analysis. The defense fails when the analyst uses material nonpublic information, so source records matter.

#494 · Ethics

What conduct counts as market manipulation?

Answer

Market manipulation is conduct intended to mislead participants by distorting prices or trading volume. Examples include false rumors and orders placed solely to create a false appearance of demand.

#495 · Ethics

Why does fair dealing not mean identical treatment?

Answer

Clients may have different objectives and may choose disclosed service levels. Fair dealing means no favored client receives an unfair timing or allocation advantage and suitable clients share opportunities through documented methods.

#496 · Ethics

What evidence is needed before judging a security suitable for a client?

Answer

The professional must understand the client's experience, return needs, risk tolerance, and financial constraints, then update them regularly. The security should fit the written objectives and constraints and be judged within the client's total portfolio.

#497 · Ethics

When may client information be disclosed?

Answer

Client information can be disclosed when the client permits it, when law requires it, or when the information concerns illegal activity. Otherwise, current, former, and prospective client information remains confidential.

#498 · Ethics

What are the main duties to an employer and a supervisor?

Answer

Professionals must act for the employer's benefit and obtain written consent from all parties for competing outside compensation. Supervisors must establish reasonable procedures, train and monitor staff, and investigate and limit possible violations promptly.

#499 · Ethics

What does full and fair conflict disclosure require?

Answer

The disclosure must be prominent, understandable, and effective before affected clients make a decision. Client and employer transactions also take priority over personal transactions, and referral benefits paid or received must be disclosed as appropriate.

#500 · Ethics

How may a CMT candidate refer to program status?

Answer

A candidate must name the exact level, such as CMT Level II Exam Candidate. A person who passes Level II may identify as a Level III candidate only while currently enrolled and eligible, or after completing the program while awaiting the designation. Only a charterholder may use the CMT credential.